



Imaging protocols for arrays of single neutral atoms

Borja Álvarez Reguera

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Advisors
Manuel Fernández Veiga
Daniel Barredo & Vincent Lienhard

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Beyond a certain point there is no return..
This point has to be reached.
— Franz Kafka

To my friends, my partner and my family.

Abstract

Abstract

This master's thesis presents the design and optimization of a high-efficiency fluorescence imaging pipeline for ^{87}Rb neutral-atom experiments. Achieving high-fidelity readout in optical tweezer arrays requires maximizing the Signal-to-Noise Ratio (SNR) within short exposure times to minimize atomic heating. The methodology employs a hybrid computational approach: Zemax OpticStudio was integrated alongside a custom Python-based grid-search algorithm to enforce proper ray tracing filters and systematically evaluate candidate lens configurations from commercial catalogs, filtering thousands of optical configurations based on magnification, track length, and pupil filling constraints. The resulting best configurations were further modeled in Zemax to perform rigorous physical optics simulations. This analysis reveals that standard spherical optics suffer from severe aberrations, identifying a symmetric aspheric relay (ASP-2) as the optimal architecture for maximizing optical performance across the whole field of view.

Regarding detection, a photon budget was established by modeling trap-induced AC Stark shifts, yielding ~ 800 incident photons per 10 ms exposure. Comparative modeling demonstrates that quantitative CMOS (qCMOS) technology significantly outperforms CCD and EMCCD sensors. By combining sub-electron read noise with a unity excess noise factor, qCMOS maintains $> 95\%$ efficiency relative to the Standard Quantum Limit, ensuring robust state discriminability even under magnification-induced spatial spreading. These findings offer a framework for evaluating readout optics, paving the way toward scalable high-fidelity detection in neutral-atom systems.

Key words: Neutral atoms, Rubidium-87, Fluorescence imaging, Ansys zemax OpticStudio, qCMOS, Signal-to-Noise Ratio (SNR).

Resumen

Esta tesis de máster presenta el diseño y la optimización de un sistema de imágenes de fluorescencia de alta eficiencia para experimentos de átomos neutros de ^{87}Rb . Lograr una lectura de alta fidelidad en matrices de pinzas ópticas requiere maximizar la relación señal-ruido (SNR) en tiempos de exposición cortos para minimizar el calentamiento atómico. La metodología emplea un enfoque computacional híbrido: se integró Zemax OpticStudio con un algoritmo de búsqueda en malla personalizado en Python para aplicar filtros adecuados de trazado de rayos y evaluar sistemáticamente configuraciones de lentes candidatas de catálogos comerciales, filtrando miles de configuraciones ópticas en función de la magnificación, la longitud del camino óptico y las restricciones de llenado de la pupila. Las mejores configuraciones resultantes se modelaron en Zemax para realizar simulaciones rigurosas de óptica física. Este análisis revela que la óptica esférica estándar

sufre aberraciones severas e identifica un sistema de relé esférico simétrico (ASP-2) como la arquitectura óptima para maximizar el rendimiento óptico en todo el campo de visión. En cuanto a la detección, se estableció un número de fotones modelando los desplazamientos AC Stark inducidos por la trampa, lo que dio como resultado ~ 800 fotones incidentes por cada 10 ms de exposición. El modelado comparativo demuestra que la tecnología CMOS cuantitativa (qCMOS) supera significativamente a los sensores CCD y EMCCD. Al combinar un ruido de lectura inferior a un electrón con un factor de exceso de ruido unitario, el qCMOS mantiene una eficiencia $> 95\%$ respecto al Límite Cuántico Estándar, asegurando una discriminación de estados robusta incluso bajo la dispersión espacial inducida por la magnificación. Estos hallazgos ofrecen un marco para evaluar la óptica de lectura, allanando el camino hacia una detección escalar de alta fidelidad en sistemas de átomos neutros.

Palabras clave: Átomos neutros, Rubidio-87, Imágenes de fluorescencia, Ansys Zemax OpticStudio, qCMOS, Relación Señal-Ruido (SNR).

Resumo

Esta tese de mestrado presenta o deseño e a optimización dun sistema de imaxes de fluorescencia de alta eficiencia para experimentos de átomos neutros de ^{87}Rb . Acadar unha lectura de alta fidelidade en matrices de pinzas ópticas require maximizar a relación sinal-ruído (SNR) en tempos de exposición curtos para minimizar o quecemento atómico. A metodoloxía emprega un enfoque computacional híbrido: integrouse Zemax OpticStudio xunto con un algoritmo de busca en malla personalizado en Python para aplicar filtros axeitados de trazado de raios e avaliar sistematicamente configuracións de lentes candidatas de catálogos comerciais, filtrando miles de configuracións ópticas en función da magnificación, a lonxitude do camiño óptico e as restricións de checado da pupila. As melloras configuracións resultantes modeláronse en Zemax para realizar simulacións rigorosas de óptica física. Este análise revela que a óptica esférica estándar sofre aberracións severas e identifica un sistema de relé esférico simétrico (ASP-2) como a arquitectura óptima para maximizar o rendemento óptico en todo o campo de visión.

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Palabras chave: Átomos neutros, Rubidio-87, Imaxes de fluorescencia, Ansys Zemax OpticStudio, qCMOS, Relación Sinal-Ruído (SNR).

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1 Rydberg atoms: fundamentals and state of the art

1.1 Introduction to Rydberg physics

After being studied extensively throughout the 20th century, research into Rydberg physics has experienced a major revival in recent years. Driven by potential applications in quantum information processing, quantum simulation, and fundamental physics, interest in high-lying atomic excitations sparked anew at the beginning of the 21st century [1]. Nowadays, the study and utilization of Rydberg states extend across a broad spectrum of physical systems: from cold [2, 3] and ultracold atoms [4, 5, 6] to thermal hot vapors [7], trapped ions [8], and even excitons in semiconductor crystals [9].

By definition, Rydberg atoms are atomic species in electronically highly excited states, characterized by a *single valence electron promoted to a large principal quantum number* ($n \gg 1$, typically in the range $30 < n < 200$). These systems are commonly realized using neutral alkali metals—such as rubidium or cesium—belonging to the first group of the periodic table. In these species, the inner-shell electrons screen the nuclear charge, allowing the outermost valence electron to experience an effective single core charge and behave similarly to a hydrogen atom.

Building upon the empirical spectral formulations by Balmer [10] and Rydberg [11], Niels Bohr established the theoretical expression for the binding energy of hydrogenic states [12]:

$$E_n = -\frac{hcR_\infty}{n^2}, \quad (1.1)$$

where h is Planck's constant, c is the speed of light, and $R_\infty = \frac{m_e e^4}{8\varepsilon_0^2 h^3 c}$ is the Rydberg constant defined purely via fundamental constants (m_e being the electron mass, e the elementary charge, and ε_0 the vacuum permittivity). For alkali atoms, low angular momentum orbitals ($l \leq 3$) penetrate the inner core shells, experiencing an increased effective nuclear charge. To account for this core penetration, Quantum Defect Theory [13, 14] modifies the energy spectrum using an effective principal quantum number $n^* = n - \delta_l$:

$$E = -\frac{hcR^*}{(n^*)^2}, \quad (1.2)$$

where δ_l is the state-dependent quantum defect (measured precisely via millimeter-wave spectroscopy [15, 16]) and R^* is the reduced mass Rydberg constant.

As a consequence of the large spatial extent of the valence electron orbital, key physical properties follow steep universal scaling laws with n^* , as summarized in Table 1.1. Notably, the huge transition dipole matrix elements and polarizabilities render these atoms extremely sensitive to external fields, while their long radiative lifetimes ($\sim 100 \mu\text{s}$) provide a extended coherence window relative to typical experimental interaction times.

Property	Expression	Scaling
Binding energy	E_n	$(n^*)^{-2}$
Orbital radius	$\langle r \rangle$	$(n^*)^2$
Radiative lifetime	τ	$(n^*)^3$
Electric polarizability	α	$(n^*)^7$
Transition dipole moment	$\langle n'l' d nl \rangle$	$(n^*)^{-3/2}$
Energy level spacing	ΔE_n	$(n^*)^{-3}$
Resonant dipole-dipole coupling	C_3	$(n^*)^4$
van der Waals coefficient	C_6	$(n^*)^{11}$

Table 1.1: Universal scaling laws for fundamental properties of Rydberg atoms as a function of the effective principal quantum number n^* [17].

Because of these exaggerated properties, Rydberg atoms provide exceptional control over interatomic forces. The colossal dipole-dipole and van der Waals interactions allow experimentalists to implement robust, laser-mediated state couplings capable of encoding physical quantum bits (qubits) and generating multi-particle entanglement. Consequently, cold neutral atoms excited to Rydberg levels have emerged as a versatile candidate spanning all main pillars of quantum technologies—encompassing quantum computation, quantum simulation, and high-precision quantum sensing—as detailed throughout the state-of-the-art overview in the following section.

1.2 State of the art in neutral-atom technologies

The field of neutral atom quantum technologies has experienced exponential progress over the last fifteen years, evolving from fundamental spectroscopic studies into a leading hardware paradigm for quantum information processing, quantum simulation, and high-precision sensing [18, 19]. This progress relies on the concurrent development of fundamental quantum operations, scalable spatial trapping techniques, and enabling hardware across the full technology stack, as summarized in Fig. 1.1. In this work, we focus particularly on optical tweezers-based platforms, which impose specific optical requirements on fluorescence imaging systems to achieve single-atom resolution and high-fidelity readout.

1.2.1 Quantum computation and simulation

In digital quantum processing, neutral atom platforms have evolved from initial multi-qubit algorithm demonstrations relying on static local addressing [20, 21] to modern dynamically reconfigurable architectures capable of fault-tolerant operations. The implementation of coherent atom shuttling using optical tweezers enabled non-local entangling

1.2 State of the art in neutral-atom technologies

gates, logical qubit encoding, and all-to-all connectivity [22], ultimately leading to the first demonstrations of fault-tolerant quantum computation below the error-correction threshold [23].

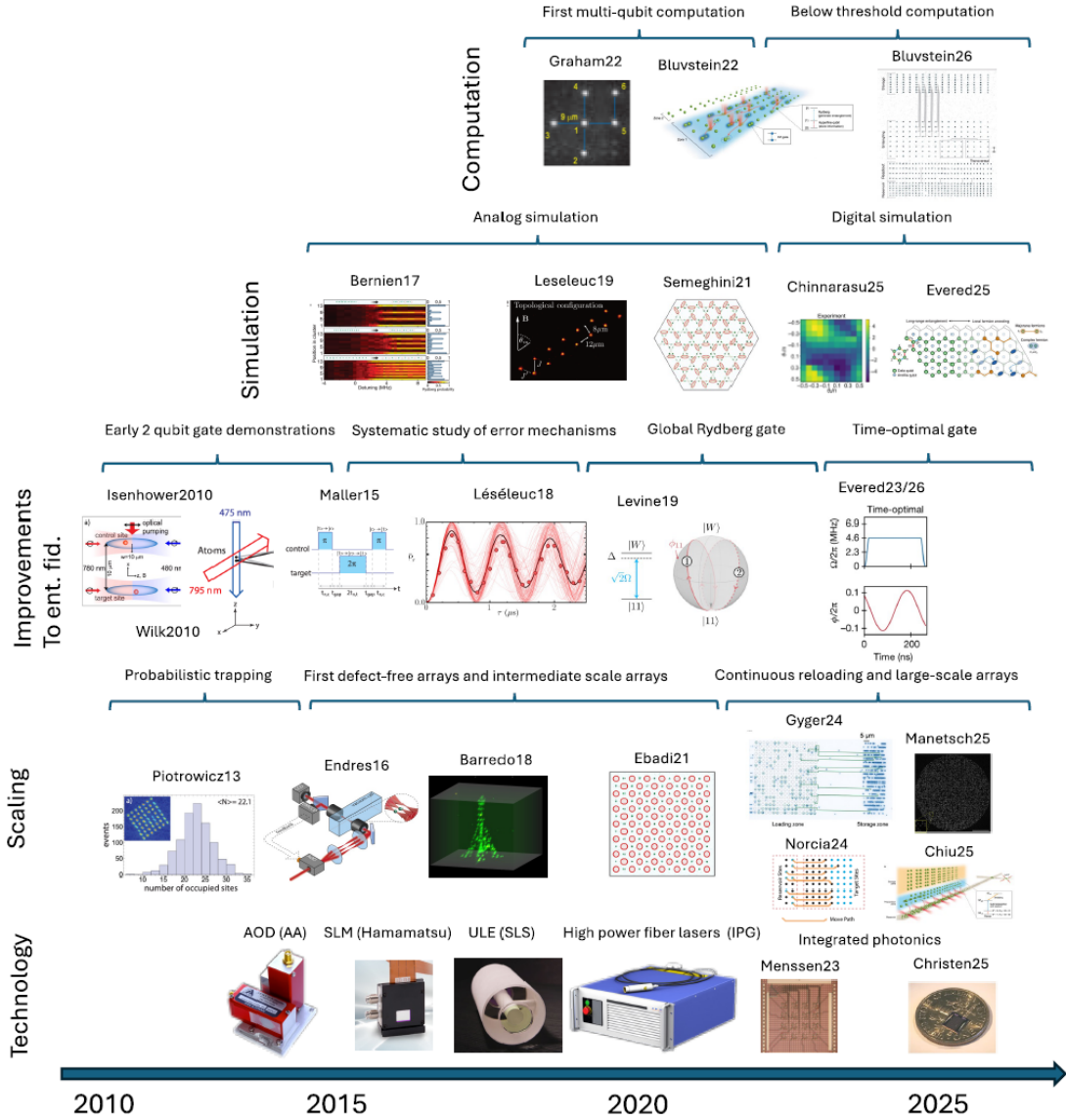


Figure 1.1: Historical and technological evolution of optical-tweezer based, and individual atom platforms. The timeline illustrates the rapid transition from fundamental two-qubit gates and probabilistic trapping (bottom left) to modern large-scale, reconfigurable architectures (bottom right). The vertical tiers categorize the parallel progress in digital computation, analog simulation, fidelity optimization protocols, and the underlying hardware technology stack, highlighting the convergence toward fault-tolerant regimes and arrays exceeding thousands of physical traps [24].

In parallel, quantum simulation has progressed from early analog implementations probing strongly interacting spin models, critical dynamics, and phase transitions [25, 26, 27, 28] toward studying exotic topological phases of matter [29]. Modern platforms have reached

a convergence where complex digital quantum simulations and combinatorial optimization problems can be executed via discrete entangling sequences over reconfigurable spatial geometries [30, 31, 32].

1.2.2 Entanglement fidelity and scaling mechanisms

The advancement in computational power has been backed by drastic improvements in entangling gate performance. Early two-qubit operations mediated by the Rydberg blockade mechanism [33, 34] were limited by technical laser noise and atomic thermal motion. To overcome these limitations, research progressed along two main avenues: detailed studies characterizing underlying error mechanisms [35, 36], and the development of advanced gate protocols, including global excitation schemes [37] and time-optimal pulse control [38, 39]. Together, these parallel efforts pushed gate fidelities beyond the stringent thresholds required for fault tolerance.

In terms of physical system size, scaling has shifted from early probabilistic optical lattice filling [40] to real-time feedback-assisted assembly of defect-free array architectures using optical tweezers in 1D, 2D, and complex 3D arrangements [41, 42, 43, 44]. Current cutting-edge tweezer architectures address atomic loss during gate sequences by integrating continuous reloading zones and architectural reservoir management [45, 46, 47, 48], ensuring long-term steady-state computation over thousands of physical traps.

1.2.3 Sensing, metrology, and quantum communications

Beyond computation and digital processing, neutral cold atoms and Rydberg states play a pivotal role in high-precision sensing and communication technologies. Neutral atom arrays held in optical tweezers or optical lattices constitute the operational foundation of state-of-the-art optical lattice clocks [49]. Coupling highly polarizable Rydberg levels to these systems offers unprecedented sensitivity in measuring gravitational shifts [50], electric field gradients [51], and fundamental physical constants [52]. Furthermore, owing to their enormous transition dipole moments, Rydberg atoms act as self-calibrated, highly sensitive sensors for radio-frequency and microwave electromagnetic fields. This property has enabled the development of atomic RF receivers and novel antennas capable of detecting weak signals across broad microwave bands with a dynamic range and noise immunity that far surpass conventional electronic sensors [53, 54].

1.2.4 Technology stack and commercial landscape

The rapid translation of cold atom physics into robust, scalable quantum hardware has been heavily driven by a mature optical and optomechanical technology stack. Dynamic beam control relies on Acousto-Optic Deflectors for rapid, high-speed atom shuttling, complemented by Spatial Light Modulators capable of projecting arbitrary holographic trapping geometries. Coherent gate manipulation requires sub-hertz laser linewidth stabilization, achieved via Ultra-Low Expansion optical cavities and complemented by high-power, low-noise fiber amplifier systems. To surpass the physical limitations of free-space optics, recent developments focus on miniaturizing these setups into integrated photonic circuits and modulator chips [55, 56], opening the path toward scaling optical

delivery to tens of thousands of individual atomic traps.

This technical standardization has fostered a thriving commercial ecosystem. Commercial entities are actively delivering cold-atom quantum processors, optical clocks, and specialized hardware: companies like QuEra Computing, Pasqal, Inflektion, and Atom Computing lead the commercialization of neutral-atom quantum processors and quantum software, while specialized vendors supply dedicated hardware modules (e.g., ColdQuanta/Inflektion for ultra-high vacuum cell subsystems and optical clocks). This industrial momentum ensures that cold-atom quantum technologies continue to transition rapidly from university research laboratories into practical, scalable industrial solutions.

1.3 Experimental platform

The experimental platform in development in El Entrego is designed to trap, manipulate, and interrogate individual neutral atoms in scalable arrays of optical tweezers. The experimental cycle begins by creating a cold atomic cloud of rubidium-87 (^{87}Rb) inside a high-vacuum chamber using a Magneto-Optical Trap (MOT). The MOT operates through the intersection of cooling laser beams and a magnetic field gradient, successfully cooling and trapping the atoms down to hundreds of microkelvin temperatures. This dense cloud serves as the continuous source from which individual traps are loaded, laying the groundwork for all subsequent quantum state manipulation and measurement procedures.

From this atomic cloud, individual atoms are captured within microscopic optical tweezers formed by focusing a trapping laser beam at 850 nm down to a sub-micrometer waist using a high-numerical-aperture objective (in the case of the figure, it is an under-vacuum asphere lens) as shown in Fig. 1.2. To scale this capability beyond a single trap, holographic techniques are implemented by placing a Spatial Light Modulator (SLM) in the optical path. This device imprints a computer-calculated phase mask onto the beam, allowing us to duplicate the single trap into an arbitrary, user-defined matrix of traps with a *typical inter-trap spacing of around 5 to 10 μm* .

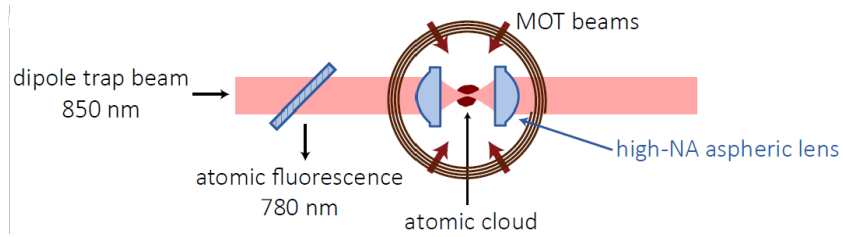


Figure 1.2: Basic elements for single-atom trapping: A cloud of cold ^{87}Rb atoms is formed in a magneto-optical trap (MOT), where a dipole trap beam is strongly focused to a $\sim 1 \mu\text{m}$ -size by a high-NA optical system (aspheric lenses in this scheme) [57].

A cornerstone of this preparation stage is the collisional blockade regime. When the spatial scale of the dipole trap is reduced to the micrometer range, light-assisted inelastic collisions occur whenever a second atom enters an already occupied trap, instantly expelling both due to the large kinetic energy gained in the process. Consequently, the occupancy of each microscopic trap is strictly regulated to either zero or one atom, yielding a 50%

loading probability per site that acts as the initial stochastic single-atom source prior to any rearrangement protocols.

Once the atomic array is prepared, quantum operations are performed by switching off the trapping beams and applying precision laser pulses to excite the atoms to high-lying Rydberg states. To maintain state purity and define a quantization axis, a stable magnetic field environment is maintained via magnetic coils. Since Rydberg atoms experience exaggerated dipole-dipole interactions, these laser excitations enable the execution of conditional logic gates and quantum simulation protocols.

The final part of the experimental sequence centers on the readout stage, which constitutes *the primary focus of this master's thesis*. To determine the internal state of each atom, state-selective fluorescence imaging is performed by illuminating the array with resonant cooling beams at 780 nm in conjunction with a repumper laser. Under this light, atoms in the target state undergo continuous absorption-emission cycles, scattering millions of photons that are collected through a high-NA objective system, separated from the trapping beams via a dichroic mirror (as shown in Fig. 1.2), and projected onto a sensitive camera. Conversely, atoms in non-target states—such as those transferred to Rydberg or decoupled hyperfine states—remain unperturbed by the resonant light or are blown away from the trap due to unbalanced photon recoil momentum and anti-trapping potentials during the detection sequence. This physical contrast maps state-selective quantum information directly into binary fluorescence signals (bright versus dark sites) across the entire array.

1.4 Fluorescence imaging and state readout

The foundational mechanism underlying the detection and readout of this quantum platform relies on collecting the atomic fluorescence emitted by the atoms. When the atoms are illuminated by the cooling laser beams, they undergo continuous optical excitation and spontaneous decay cycles associated with the D2 line [58]. Specifically, the atoms transition between the ground-state hyperfine manifolds and the $5P_{3/2}$ excited state, repeatedly scattering photons at a wavelength of 780 nm. Because this spontaneous emission is isotropic, the trapped atoms act as point sources radiating light in all directions, making the collection efficiency of our optical system the critical bottleneck for high-fidelity detection.

Imaging this fluorescence to determine trap occupancy is achieved through a dedicated optical pipeline designed and optimized in Chap. 2. The isotropic 780 nm photons emitted by the atoms are gathered by the high-numerical-aperture objective and routed through a series of optical elements engineered to preserve the intrinsic spatial resolution of the objective across the full field of view. The light is ultimately focused onto a high-performance quantitative Complementary Metal-Oxide-Semiconductor (qCMOS) camera—specifically the Hamamatsu Orca-Quest 2 model, whose integration, characteristics, and performance are thoroughly detailed in Chap. 3. Featuring ultra-low read noise and exceptional quantum efficiency, this camera allows us to directly map the spatial distribution of the trap array. By analyzing the intensity spots on the camera sensor, we can unambiguously discern whether a given optical tweezer contains a single atom (registered as a distinct fluorescence signal) or is empty (background noise level).

This imaging capability forms the core of our experimental readout scheme, requiring both spatial fidelity and accurate state discrimination. On one hand, the optical system must resolve every optical tweezer across the entire field of view—demanding high transmission for large off-axis angles and minimal Point Spread Function (PSF) spreading to satisfy the Rayleigh criterion for adjacent traps. On the other hand, this capability enables mapping internal atomic states onto a straightforward binary outcome. The experimental sequence follows a clear progression: first, individual atoms are trapped from the cold atomic cloud in the ground state (logic 0); second, they undergo the main experimental routine—such as applying a sequence of quantum gates or allowing the system to evolve under a target Hamiltonian; and third, the final state is measured. Because the detection process evaluates whether an atom remains trapped, *the physical presence of an atom in the tweezer after the sequence confirms that it ended up in the ground state*, whereas its absence indicates that it transitioned to the excited state (logic 1) and was consequently lost.

While destructive state-to-loss detection is robust and widely implemented, non-destructive state-selective imaging protocols have been demonstrated by tailoring the polarization, detuning, and repumping light to preserve atomic survival across cycles [59]. Although non-destructive readout represents a promising avenue for real-time feedback and quantum trajectory tracking, its implementation requires dedicated optical configurations that lie beyond the primary scope of this work.

In conclusion, the neutral atom platform in optical tweezers establishes itself as a versatile architecture for quantum information processing, where the ability to interrogate the atomic state via fluorescence constitutes the fundamental pillar of the readout process. However, the fidelity of this measurement is intrinsically linked to photon collection efficiency and the optical quality of the image projected onto the sensor. Having established the state preparation routines and detection mechanisms in this chapter, the following block addresses the engineering challenge of designing and optimizing the optical system required to channel this signal accurately and without aberrations.

2 Optical stage: fluorescence system design and optimization

This chapter details the design, modeling, and performance evaluation of the optical architecture dedicated to the collection and imaging of atomic fluorescence. Starting from fundamental resolution and spatial sampling constraints, we establish the necessity of a pupil-conjugated relay system to prevent off-axis vignetting. We then define the quantitative metrics used to assess image quality, followed by a formal treatment of the *étendue optique*. Finally, through a systematic analysis of lens filling factors, focal lengths, and geometric aberrations, we evaluate candidate optical configurations and select the optimal setup for good optical resolution.

2.1 Optical architecture and pupil conjugation

The fluorescence imaging line maps the spatial distribution of light emitted by the cold atomic array onto a qCMOS camera sensor. This camera consists of a two-dimensional grid of independent pixels, where each pixel acts as an individual photo-detector recording the local light intensity. Provided the imaging optics preserve the spatial coherence and resolution, this layout records the original spatial pattern of the fluorescence. In our experimental setup, the primary microscope objective that collects this isotropic fluorescence directly in front of the vacuum chamber is treated as a fixed component whose properties cannot be modified. *The design freedom is strictly limited to selecting and positioning the secondary optical elements located between this objective and the camera.*

To design this secondary stage, a first intuitive approach is to map the image of a point source (the single atom)—whose size is dictated by the diffraction-limited Point Spread Function (PSF) of the primary objective, with a characteristic width of $\sim 1 \mu\text{m}$ —onto a single camera pixel, which has a physical pitch of $4.6 \mu\text{m}$. Under this initial approximation, the required optical system magnification is estimated to be $M \sim 4.6$.

Achieving this magnification could ideally be accomplished using a single tube lens combined with the fixed microscope objective. The objective itself is characterized by key parameters that define its light-gathering and field limits: its Numerical Aperture ($\text{NA} = n \sin \theta = 0.65$), which represents the half-angle θ of the collected light cone; its Effective Focal Length (f_{obj}); and its Field of View ($\text{FoV} = 1.5 \text{ mm}$), which specifies the maximum spatial region around the optical axis where geometric aberrations remain controlled. While a two-lens arrangement (objective plus a single tube lens) could provide

2.1 Optical architecture and pupil conjugation

the target magnification $M = f_{\text{tube}}/f_{\text{obj}}$ for emitters located exactly on the optical axis, it fails for emitters distributed across the extended FoV due to severe field aberrations and optical vignetting, which clips off-axis light cones and reduces collection efficiency.

For off-axis atoms, light rays emerge from the objective at non-zero propagation angles. In a simple single-tube-lens setup, as illustrated in Fig. 2.1a, the exit pupil of the objective is not projected onto subsequent optical elements. As a result, the light cones originating from off-axis emitters walk off the clear aperture of the tube lens, leading to severe beam clipping and field-dependent vignetting toward the edges of the sensor. To catch all the light emitted across the maximum field of view (FoV = 1.5 mm), the required clear aperture diameter (D_{lens}) of a single tube lens placed at a distance d from the objective exit pupil would need to satisfy $D_{\text{lens}} \geq D_{\text{pupil}} + 2d \tan \theta_{\text{max}}$, where $D_{\text{pupil}} = 2 \text{NA} \cdot f_{\text{obj}}$ is the exit pupil diameter and $\theta_{\text{max}} = \arctan(\text{FoV}/2f_{\text{obj}})$ represents the maximum field angle. For typical focal distances, this condition demands an unrealistically large lens diameter, making a single-tube-lens approach unfeasible.

To mitigate these field-dependent losses, the optical train implements *pupil conjugation* using a pair of relay lenses positioned between the objective and the final tube lens (see Fig. 2.1b). Beyond correcting vignetting, this multi-stage relay design is also mechanically required, as the physical clearance around the vacuum chamber and the placement of optical interference filters prevent positioning a single tube lens close enough to the objective. Through this intermediate telescope, the objective exit pupil—located inside the objective housing with a diameter $D_{\text{pupil}} = 2 \text{NA} \cdot f_{\text{obj}}$ —is re-imaged onto a conjugated pupil plane near the final tube lens. By controlling the chief ray trajectories through this relay system, off-axis light cones are guided through the clear apertures of the secondary optics without encountering the severe edge clipping typical of a single-lens setup. Provided the clear apertures of the relay lenses are sized to accommodate the field-dependent beam displacement, this conjugated arrangement significantly mitigates vignetting, leading to a much more uniform collection efficiency across the full field of view.

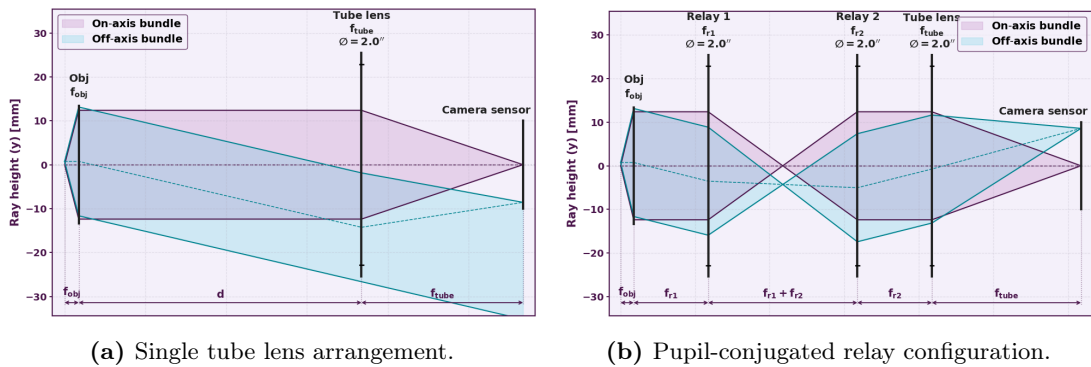


Figure 2.1: Schematic representation of the fluorescence collection systems. (a) A single tube lens arrangement showing vignetting and aperture walk-off for off-axis emitters. (b) Insertion of two additional lenses forming a 4f relay telescope between the objective and the tube lens to maintain pupil alignment and uniform ray collection across the full Field of View.

2.2 *Étendue optique* formalism and optical requirements

2.2.1 Definition and conservation of the *étendue optique*

The fixed parameters of the microscope objective impose fundamental geometric bounds on the entire optical setup. The displacement of a field point in the atomic plane x_{field} relates to the incidence angle through the objective focal length f_{obj} , which yields the maximum angle required to reach the edge of the field of view:

$$x_{\text{field}} = f_{\text{obj}} \cdot \theta_{\text{in}} \Rightarrow \theta_{\text{in}}^{\text{max}} = \frac{\text{FoV}/2}{f_{\text{obj}}}. \quad (2.1)$$

Simultaneously, the physical beam size (or pupil diameter) required to achieve the nominal numerical aperture is given by $D = 2 \cdot \text{NA} \cdot f_{\text{obj}}$. Multiplying D by the maximum angular spread $\theta_{\text{in}}^{\text{max}}$ yields the *étendue optique* (G):

$$G = D \cdot \theta_{\text{in}}^{\text{max}} = (2 \cdot \text{NA} \cdot f_{\text{obj}}) \cdot \left(\frac{\text{FoV}/2}{f_{\text{obj}}} \right) = \text{FoV} \cdot \text{NA}. \quad (2.2)$$

As shown in Eq. (2.2), the focal length f_{obj} cancels out completely, confirming that G is a *fundamental phase-space invariant conserved across any ideal, lossless optical train*.

We can interpret this invariant through diffraction and information theory. The maximum number of independent spatial modes (or distinct optical traps) N_{modes} that can be packed into our field of view is determined by dividing the total field size by the minimum resolvable separation distance Δx_{min} . Whether we define this threshold using the Gaussian waist radius ($\Delta x_{\text{min}} = \omega_0 = \lambda/\pi \cdot \text{NA}$), the full beam diameter ($\Delta x_{\text{min}} = 2\omega_0$), or the classical Rayleigh criterion ($\Delta x_{\text{min}} = 0.61 \cdot \lambda/\text{NA}$) for circular pupils, the structural scaling law remains invariant:

$$N_{\text{modes}} = \frac{\text{FoV}}{\Delta x_{\text{min}}} \propto \frac{\text{FoV} \cdot \text{NA}}{\lambda} = \frac{G}{\lambda}. \quad (2.3)$$

The *étendue optique*, normalized by the wavelength, defines the absolute information capacity of the system. It dictates how many independent spatial channels can be resolved before diffraction causes neighboring optical traps to overlap or causes light collected from adjacent atoms to irreversibly blend together, drawing a direct parallel to the Nyquist-Shannon sampling theorem in signal processing [60, 61].

2.2.2 Trapping line constraints and magnification trade-offs

The optical trapping line—defined as the optical subsystem responsible for shaping, steering, and projecting the off-resonant laser beams used to create the optical tweezers—is designed to project dynamic, high-resolution light patterns onto the atomic plane to generate the dipole trap array. A laser beam at $\lambda_{\text{trap}} = 850 \text{ nm}$ illuminates a pixelated Spatial Light Modulator (SLM), which imparts a calculated spatial phase pattern. This phase profile acts as a digital hologram, wavefront corrector, and focusing element to sculpt the desired intensity landscape. A downstream relay telescope subsequently images the illuminated SLM active area onto the back focal plane of the microscope objective, enforcing a strict pupil conjugation that maps the spatial phase modulations into positions of focused optical tweezers at the focal region.

2.2 *Étendue optique* formalism and optical requirements

In this architecture, the *pixelated nature of the SLM introduces a fundamental input phase-space constraint*. To avoid digital aliasing, the spatial Nyquist criterion dictates a minimum period of two pixels ($2p$, with pitch $p = 12.5 \mu\text{m}$), bounding the maximum clean deflection angle at $\lambda_{\text{trap}} = 850 \text{ nm}$ to $\theta_{\text{SLM}}^{\text{max}} = \arcsin \lambda_{\text{trap}}/2p \simeq 34 \text{ mrad}$. Illuminating a diameter $D_{\text{SLM}} \simeq 12.35 \text{ mm}$ injects a fixed input *étendue optique* of $G_{\text{trap}} = D_{\text{SLM}} \cdot \theta_{\text{SLM}}^{\text{max}} \simeq 0.42 \text{ mm}$. Due to this strict conservation along the optical path, G_{trap} establishes an upper limit on the available phase-space, generating effective field sizes (FoV_{eff}) and numerical apertures (NA_{eff}) at the atomic plane that are strictly lower than the nominal upper limits of the microscope objective.

These effective parameters remain bound by the hyperbola $G_{\text{trap}} = \text{FoV}_{\text{eff}} \cdot \text{NA}_{\text{eff}}$. Adjusting the magnification M of the relay telescope positioned downstream of the SLM transforms them according to:

$$\text{FoV}_{\text{eff}} = \frac{2 \cdot f_{\text{obj}} \cdot \theta_{\text{SLM}}^{\text{max}}}{M}, \quad \text{NA}_{\text{eff}} \simeq \frac{D_{\text{SLM}} \cdot M}{2 \cdot f_{\text{obj}}}. \quad (2.4)$$

This trade-off directly affects the physical geometry and confinement of individual optical tweezers, as shown in Eq. (2.5):

$$\omega_0 \simeq \frac{\lambda_{\text{trap}}}{\pi \cdot \text{NA}_{\text{eff}}}, \quad U_0 \propto \frac{P}{\omega_0^2} \propto P \cdot \text{NA}_{\text{eff}}^2. \quad (2.5)$$

where ω_0 is the Gaussian beam waist radius at the focal plane, U_0 is the resulting dipole trap depth, and P is the optical power per trap. Demagnifying the relay optics ($M < 1$) to achieve a larger array area severely compromises the trap depth, requiring significantly higher laser power to maintain atomic confinement. Fig. 2.2 maps this phase-space trade-off across values of FoV_{eff} , NA_{eff} , ω_0 and M .

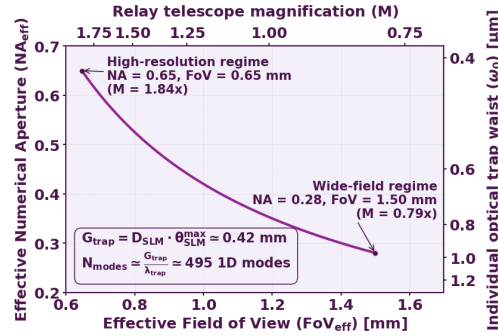


Figure 2.2: Phase-space trade-off between effective Field of View (FoV_{eff}), effective Numerical Aperture (NA_{eff}), and trap waist radius (ω_0) dictated by the *étendue optique* conservation ($G_{\text{trap}} \simeq 0.42 \text{ mm}$). Tuning the relay magnification M balances the accessible array area against the trap waist size and depth.

2.2.3 Fluorescence line bounds and automated lenses selection

The fluorescence imaging line operates in the inverse direction of the trapping mechanism. The system collects isotropic photon emission at $\lambda_{\text{fluor}} = 780 \text{ nm}$ from the ^{87}Rb atoms to

perform high-fidelity binary readout—determining whether a trapping site is occupied or empty—rather than resolving internal trap profiles. To maximize the signal-to-noise ratio against camera readout noise within short exposure windows, the objective collects the maximum numerical aperture ($\text{NA} = 0.65$) across the full field of view ($\text{FoV} = 1.5 \text{ mm}$). This establishes a maximum baseline fluorescence *étendue optique* of:

$$G_{\text{fluo}} = \text{FoV} \cdot \text{NA} = 1.5 \text{ mm} \cdot 0.65 = 0.975 \text{ mm}. \quad (2.6)$$

Matching G_{fluo} to the active area of a qCMOS sensor (Hamamatsu ORCA-Quest 2, pixel pitch $p_{\text{sensor}} = 4.6 \mu\text{m}$) imposes two strict spatial bounds on the total system magnification M_{total} :

$$M_{\text{min}} > \frac{p_{\text{sensor}}}{\Delta x_{\text{atoms}}}, \quad M_{\text{max}} < \frac{\min\{x_{\text{sensor}}, y_{\text{sensor}}\}}{\text{FoV}}; \quad (2.7)$$

where $\Delta x_{\text{atoms}} \simeq 5 - 10 \mu\text{m}$ is the physical spatial separation between adjacent atomic trapping sites, and $x_{\text{sensor}} = 18.841 \text{ mm}$, $y_{\text{sensor}} = 10.598 \text{ mm}$ denote the active physical dimensions of the sensor matrix.

The lower bound M_{min} guarantees single-site spatial resolution. To perform unambiguous binary occupancy readout, the geometric separation between the centers of two adjacent atoms projected onto the camera plane ($M_{\text{total}} \cdot \Delta x_{\text{atoms}}$) must strictly exceed the physical pixel pitch p_{sensor} , provided that the point spread function (PSF) spread of the optical system is sufficiently small. This prevents the central intensity lobes of neighboring traps from falling onto the same pixel coordinate and minimizes optical signal crosstalk. Conversely, the upper bound M_{max} sets a geometric enclosure limit: expanding the image beyond this threshold causes the outer regions of the full Field of View ($\text{FoV} = 1.5 \text{ mm}$) to extend past the physical bounds of the sensor chip, leading to spatial clipping and loss of peripheral trap channels. Together, Eq. (2.7) defines the operational window ($M_{\text{min}}, M_{\text{max}}$) required for reliable readout.

Because mechanical clearance around the vacuum chamber and space for optical interference filters prevent using a single tube lens, the setup employs a multi-stage relay telescope. Paraxial ray tracing yields the overall system magnification as a function of the relay and tube lens focal lengths:

$$M_{\text{total}} = \frac{f_{\text{tube}}}{f_{\text{obj}} \cdot M_{\text{relay}}} = \frac{f_{\text{tube}}}{f_{\text{obj}}} \cdot \frac{f_{\text{relay1}}}{f_{\text{relay2}}}. \quad (2.8)$$

Selecting commercial catalog lenses introduces severe geometric trade-offs. Demagnifying relays ($M_{\text{relay}} < 1$) keep the central beam within standard clear apertures (typically $\text{CA} \simeq 0.9\varnothing$) but expand the angular spread of off-axis chief rays ($\theta_{\text{out}} = \theta_{\text{in}}/M_{\text{relay}}$), risking image clipping. Conversely, magnifying relays ($M_{\text{relay}} > 1$) compress the angular spread but expand the physical beam diameter, risking severe off-axis optical aberrations and vignetting at the lens edges. Furthermore, the total optomechanical track length must satisfy physical footprint constraints:

$$L_{\text{total}} = \text{WD} + L_{\text{obj}} + 2f_{\text{relay1}} + 2f_{\text{relay2}} + f_{\text{tube}}, \quad (2.9)$$

where $\text{WD} = 23 \text{ mm}$ is the objective working distance and L_{obj} is its physical housing length.

To resolve these coupled constraints, a *grid-search optimization algorithm* was implemented to evaluate every candidate combination $(f_{\text{relay1}}, f_{\text{relay2}}, f_{\text{tube}})$ across a large subset of the commercial Thorlabs lens catalog. Candidate setups are systematically evaluated through five sequential filtering stages:

- † **Magnification bounding:** Discard triplets falling outside the operational window $M_{\text{min}} \leq M_{\text{total}} \leq M_{\text{max}}$.
- † **Track length constraints:** Enforce $50 \text{ cm} \leq L_{\text{total}} \leq 200 \text{ cm}$ to guarantee optomechanical ergonomics while preventing thermal and mechanical drift.
- † **Collimated chief ray angle:** Bound off-axis propagation angles to $|\theta_{\text{coll}}| \leq 5^\circ$ in collimated spaces to limit high-order aberrations.
- † **Beam filling factor:** Bounding the total footprint $D_{\text{footprint}}$ relative to the outer lens diameter $\varnothing_{\text{lens}}$ ($D_{\text{footprint}} \leq \text{Filling Factor} \cdot \varnothing_{\text{lens}}$) to prevent peripheral clipping and aberrations.
- † **Focal length floor:** Set $f_{\text{relay1}}, f_{\text{relay2}}, f_{\text{tube}} \geq 100 \text{ mm}$ to preserve working clearances.

A comprehensive analysis evaluating how filling factor thresholds interact with different lens topologies to impact final spot quality will be presented in subsequent sections.

2.3 Performance evaluation metrics

Before proceeding further with the design and optimization workflow of the total fluorescence collection system, this section opens a brief parenthesis to define and detail the quantitative metrics used to assess focusing quality and optical aberrations across individual lenses and multi-element combinations. In optical design, *an optical aberration represents any deviation of a real optical system from ideal paraxial behavior*, causing light rays originating from a single object point to fail to converge into a diffraction-limited image spot. To quantitatively evaluate these imperfections, simulations were performed using the optical design software Ansys Zemax OpticStudio (version 2026 R1) [62]. The following subsections describe the key evaluation criteria applied throughout this study: the Huygens Point Spread Function (PSF) cross section, the Strehl ratio, the wavefront error, and the Zernike polynomial coefficients.

2.3.1 Huygens Point Spread Function and cross-section

The spatial distribution of optical power at the focal plane of an imaging system illuminated by a point source dictates its spatial resolving power and image quality. Physical optics models this through the Point Spread Function (PSF), which accounts for wave diffraction governed by the Huygens-Fresnel principle as light passes through finite optical apertures. The spatial extent of the PSF fundamentally sets the resolution limit of the system, defined by the Rayleigh criterion for resolving two adjacent point sources as:

$$r_{\text{Rayleigh}} = 0.61 \frac{\lambda}{\text{NA}}, \quad (2.10)$$

where λ is the working wavelength and NA is the numerical aperture of the objective lens. In Zemax OpticStudio, the output PSF corresponds to a two-dimensional intensity map across the image plane computed via numerical diffraction theory.

Numerical diffraction models rely on mathematical simplifications of the Optical Path Length (OPL) distance between pupil and image coordinates,

$$r_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}. \quad (2.11)$$

Standard Fast Fourier Transform (FFT) methods apply the Fraunhofer approximation, truncating the binomial expansion of r_{ij} to linear phase terms to model far-field propagation. Intermediate regimes employ the Fresnel approximation, retaining quadratic parabolic distance terms. While computationally efficient, these approximations neglect higher-order phase variations across heavily aberrated or steep wavefronts. To guarantee maximum physical accuracy, the Huygens PSF computation in Zemax OpticStudio was selected for this study. Although computationally more intensive, this method avoids Taylor series truncations altogether by evaluating the exact Euclidean distance r_{ij} across a dense grid of pupil rays, integrating their complex wave amplitudes at the detector plane.

One-dimensional cross-sections taken through the peak irradiance of the resulting 2D Huygens PSF map isolate the central diffraction peak and surrounding Airy ring structures. *From these profiles, spatial beam parameters such as the spot radius at the $1/e^2$ intensity threshold ($\sim 13.5\%$ of peak irradiance) are extracted.* While expanding the pupil filling factor increases the numerical aperture and theoretically narrows the central diffraction spot (explaining the tendency shown in Figs. 2.3a, 2.4a, 2.5a and 2.6a), it also amplifies marginal ray aberrations that broaden the main lobe and redistribute power into side halos.

2.3.2 Strehl ratio

The Strehl ratio, S , provides a single scalar metric quantifying how closely an optical system approaches ideal diffraction-limited performance. It is defined as the ratio of the peak intensity of the actual aberrated PSF to that of an ideal, aberration-free PSF under identical aperture and wavelength conditions:

$$S = \frac{I_{\max, \text{abe}}^{\text{PSF}}}{I_{\max, \text{no abe}}^{\text{PSF}}}. \quad (2.12)$$

By Maréchal's criterion, an optical system is formally considered diffraction-limited when $S \geq 0.8$. While individual relay lenses act primarily as afocal beam transfer optics where wavefront deformation or angular aberration growth are the primary concerns, the Strehl ratio becomes the decisive figure of merit when evaluating the total imaging system—specifically at the image plane formed by the final tube lens. At this terminal stage, a high Strehl ratio ensures that fluorescence photons originating from a single point-like atom are tightly focused onto the camera sensor, maximizing the peak signal-to-noise ratio and preventing signal bleed into adjacent pixels. In this work, Strehl ratio values are extracted from the physical optics Huygens PSF cross-sections in Zemax

OpticStudio. Note that Zemax documentation restricts the numerical reliability of scalar Strehl calculations to $S > 0.1$, below which severe aberrations spread energy so extensively that peak-based metrics fail to reflect true optical quality [62].

2.3.3 Wavefront error

The wavefront error maps the phase deviation of real rays across the exit pupil relative to an ideal spherical reference wave centered at the paraxial focus. This continuous Optical Path Difference (OPD) distribution is synthesized into two scalar parameters: Peak-to-Valley (P-V) error—measuring the maximum absolute phase difference—and Root-Mean-Square (RMS) wavefront error (σ), which evaluates the spatial standard deviation of the OPD over the pupil area.

To compute σ with high integration accuracy while minimizing ray tracing density, Zemax OpticStudio utilizes Gaussian quadrature over Forbes sampling nodes with integration weights w_k , evaluated via the Rimmer algorithm:

$$\sigma \equiv W_{\text{RMS}} = \sqrt{\sum_{k=1}^N w_k (W_k - \bar{W})^2}, \quad \text{where} \quad \bar{W} = \sum_{k=1}^N w_k W_k. \quad (2.13)$$

2.3.4 Zernike polynomial coefficients

While classical Born-Wolf power series express wave aberrations in terms of non-orthogonal Seidel terms over circular apertures [63], Standard Zernike polynomials $Z_i(\rho, \theta)$ form a complete, orthonormal basis over a unit circle. An arbitrary pupil phase deformation $W(\rho, \theta)$ can thus be uniquely expanded into a linear combination of these orthogonal modes weighted by scalar coefficients c_i :

$$W(\rho, \theta) = \sum_{i=0}^{\infty} c_i Z_i(\rho, \theta). \quad (2.14)$$

The primary mathematical advantage of this orthonormal basis is that the overall wavefront variance σ^2 decouples cleanly into a sum of independent, squared non-piston coefficients:

$$\sigma^2 = \sum_{i=1}^{\infty} c_i^2. \quad (2.15)$$

This property allows the total RMS error to be parsed into distinct physical aberration modes without cross-term interference. Tracking specific coefficients as a function of pupil filling factor isolates the exact individual contributions of defocus ($Z_4 \propto 2\rho^2 - 1$), primary astigmatism ($Z_5, Z_6 \propto \rho^2 \cos 2\theta$), primary coma ($Z_7, Z_8 \propto (3\rho^3 - 2\rho) \cos \theta$), and primary spherical aberration ($Z_{11} \propto 6\rho^4 - 6\rho^2 + 1$).

2.3.5 Analytical link between Strehl ratio and RMS wavefront error

The analytical bridge connecting focal plane peak irradiance to exit pupil phase variance relies on approximations valid for low-magnitude phase errors ($\sigma \ll \lambda$). Expanding the

Strehl ratio for small phase variances yields Maréchal’s classical quadratic Taylor series:

$$S \simeq 1 - (2\pi\sigma)^2 + \dots \quad (2.16)$$

A more general statistical formulation established by Ross [64] links the Strehl ratio to the Fourier transform of the phase aberration Probability Density Function (PDF), $S = \sigma^2 |\mathcal{F}[\text{PDF}(\hat{\xi})]|^2$. When the pupil phase noise follows a Gaussian distribution, this transform yields the exact exponential identity $S = e^{-\sigma_\phi^2}$ where $\sigma_\phi^2 = (2\pi\sigma)^2$ represents the phase variance in radians. Furthermore, by the Central Limit Theorem, as an optical system accumulates multiple independent surface errors, the combined phase PDF naturally converges toward a Gaussian distribution, *explaining why the exponential model remains robust even for non-Gaussian individual surface errors*.

In the high-quality regime ($S \geq 0.8$), exponential, quadratic, and sinc-squared models converge to nearly identical numerical values. Evaluating Maréchal’s criterion ($S \geq 0.8$) within this framework yields the classic diffraction-limited tolerance threshold of $\sigma \leq \lambda/14$.

2.4 Lens filling factors and focal length selection

To establish rigorous design criteria for the total fluorescence collection system, this section evaluates the focusing performance and aberration dynamics of various commercial lens geometries as a function of focal length and pupil filling factor. In the experimental setup, this filling factor varies according to the magnification of the relay stage—given the fixed beam diameter exiting the objective lens—and the physical clear aperture of the chosen optics. Furthermore, while the performance requirements for the relay lens and the tube lens distinctively diverge depending on whether intermediate collimation or camera conjugation is prioritized, this initial screening provides a unified baseline for both components. To quantify these trade-offs between spatial beam concentration and optical wavefront degradation, simulations were systematically executed in Ansys Zemax OpticStudio using both geometric ray-tracing (for marginal ray paths and classical aberrations) and physical optics propagation (accounting for diffraction and phase evolution).

To provide a focused analysis while adhering to document length constraints, the core discussions are structured around three representative comparison groups selected from a broader parametric study:

† **Lens architecture comparison:** Evaluating performance metrics across distinct optical families at a fixed focal length of $f = 100$ mm, where each geometry presents specific structural properties and expected aberration profiles [65]:

- ‡ *Paraxial reference:* Idealized, aberration-free transformation used as an analytical baseline.
- ‡ *Plano-convex singlets:* Simple spherical surfaces designed for infinite conjugate ratios, expected to introduce significant spherical aberration at small f -numbers due to non-uniform marginal ray refraction.
- ‡ *Achromatic doublets:* Multi-element cemented pairs combining crown and flint glasses, engineered to minimize spherical aberration and cancel primary chromatic dispersion across the target spectral band.

‡ *Aspheric singlets:* Non-spherical surface profiles tailored to systematically eliminate high-order spherical aberration, providing near-diffraction-limited focusing at the expense of field-angle tolerance.

† **Focal length effect:** Assessing how focal distance variations influence spherical aberration in both achromatic doublets and singlet plano-convex lenses.

† **Diameter effect:** Investigating the impact of scaling the physical lens diameter (1" vs. 2") across matching and scaled focal lengths.

Additional optical configurations—such as the systematic evaluation of lens orientation (inversion effect)—were also simulated across all candidate lenses. In agreement with classical geometrical optics, orienting the curved surface of a lens toward the incoming collimated beam distributes refraction across both interfaces, suppressing marginal ray spherical aberration (Z_{11}). Conversely, facing the planar surface to the collimated beam forces all angular deviation onto a single interface, severely degrading focal quality. For brevity, graphical cross-sections of these inverted states are omitted here; however, the complete parametric dataset, raw Zemax outputs, and extended comparison plots across all simulated groups are publicly accessible in the project repository [66].

2.4.1 Zemax simulations procedure

The simulation workflow systematically evaluated commercial single lenses under on-axis monochromatic illumination ($\lambda = 780$ nm), modeling the input beam as a perfectly collimated light source propagating parallel to the optical axis. While restricting the present analysis to on-axis propagation constitutes a simplified baseline—omitting off-axis aberrations such as coma and astigmatism that represent a known limitation of this section and are deferred to future research—it provides a clear upper bound on the performance. Initial focal distances followed manufacturer specifications from Thorlabs, incorporating wavelength-specific focal shift corrections where applicable (e.g., in achromatic doublets). For each lens, the entrance pupil diameter was expanded from 25% to 90% of the maximum clear aperture in 5% increments. At each step, an automated Quick Focus optimization repositioned the image plane to minimize the RMS wavefront error, a focal distance highly sensitive to pupil filling that shifted from micrometers to millimeters depending on surface curvature and marginal ray spherical aberration.

Preliminary test simulations established fixed high-precision parameters to guarantee numerical convergence across the full catalog. Zernike polynomial analysis used a 512×512 ray pupil grid to fit terms up to Z_{40} , while Huygens PSF calculations combined a 512×512 ray pupil sampling with a 256×256 detector array, leaving the image delta at its default setting for dynamic grid scaling. Exported PSF profiles were normalized to unity peak intensity to preserve visual readability in severe low-Strehl regimes, overcoming Zemax's default ideal-system scaling that heavily attenuates aberrated peaks. For every optical state, layout schematics, fitted Zernike coefficients, and raw/graphic Huygens PSF cross-sections were systematically exported—a procedure requiring 8–10 minutes per step (~ 2 hours per lens) and representing dozens of evaluated states.

To facilitate comparison across all parametric evaluation groups, *the resulting optical metrics are presented in a standardized two-panel figure format*. Panel (a) tracks global

focal performance metrics as a function of the pupil filling factor, including the Strehl ratio (S), RMS wavefront error (σ), Peak-to-Valley (P-V) phase distortion, and $1/e^2$ beam spot radius. Panel (b) isolates the underlying phase dynamics by plotting the expansion of rotationally symmetric Zernike polynomial coefficients (Z_4 for defocus, alongside Z_{11} , Z_{22} , and Z_{37} for primary, secondary, and tertiary spherical aberrations, respectively). These higher-order spherical terms directly govern the redistribution of power from the central Airy peak into surrounding diffractive rings, distorting the point spread function (PSF); suppressing these coefficients is critical for single-atom imaging to prevent cross-talk between adjacent trap sites and maintain high signal-to-noise fluorescence collection.

2.4.2 Lens architecture comparison

Fig. 2.3 evaluates the focal quality and Zernike expansions for all candidate $f = 100$ mm geometries across pupil filling factors from 25% to 90%.

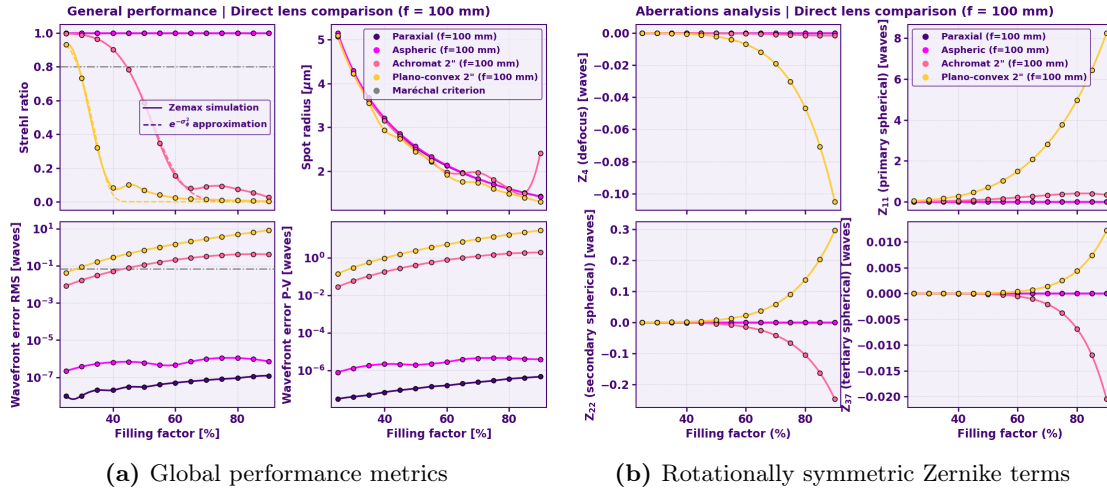


Figure 2.3: Comparative optical performance for $f=100$ mm lens architectures across pupil filling factors. The aspheric lens maintains diffraction-limited performance ($S \geq 0.8$) throughout the entire aperture, behaving ideally. In contrast, the achromatic doublet and plano-convex singlet exhibit rapid quality degradation beyond 45% and 30% filling, respectively, driven by the uncontrolled growth of primary spherical aberration.

As shown in Fig. 2.3a, overall focal quality degrades rapidly at higher filling factors due to phase distortion from marginal rays. Following Maréchal's diffraction-limit criterion ($S \geq 0.8$), the commercial aspheric lens maintains diffraction-limited performance across the entire pupil aperture, behaving virtually identically to the ideal paraxial. In contrast, the plano-convex singlet and achromatic doublet drop below this threshold at filling factors of approximately 30% and 45%, respectively. This drop in Strehl ratio was cross-validated by independently evaluating the Huygens PSF peak intensity and the RMS wavefront error (σ) in Zemax OpticStudio, confirming that the exponential decay $S \simeq \exp(-\sigma^2)$ introduced in Sec. 2.3.5 accurately reflects physical phase variance rather than a numerical artifact. Furthermore, while the nominal $1/e^2$ spot radius for singlets and doublets may remain within small diffraction limits, spot size alone is an insufficient figure of merit for single-atom imaging; uncorrected spherical aberration redirects significant power from the

central core into broad diffraction halos, lowering signal collection efficiency and causing severe optical cross-talk with neighboring atomic sites.

The Zernike expansion in Fig. 2.3b reveals the primary optical aberrations driving these trends. The precision surface profile of the aspheric lens suppresses all evaluated orders across the entire aperture range. The plano-convex singlet exhibits progressive growth across all modes (Z_4 through Z_{37}) as higher aperture zones are illuminated. The achromatic doublet successfully suppresses lower-order terms (Z_4 and Z_{11}) at moderate filling factors, but higher-order spherical contributions (Z_{22} and Z_{37}) grow substantially at large apertures, reaching magnitudes comparable to those of the simple singlet.

2.4.3 Focal length effect

To evaluate how focal distance influences wave aberration dynamics, Fig. 2.4 and Fig. 2.5 present the parametric performance across focal lengths ($f \in \{100, 150, 200\}$ mm) for 2" diameter achromatic doublets and plano-convex singlets, respectively.

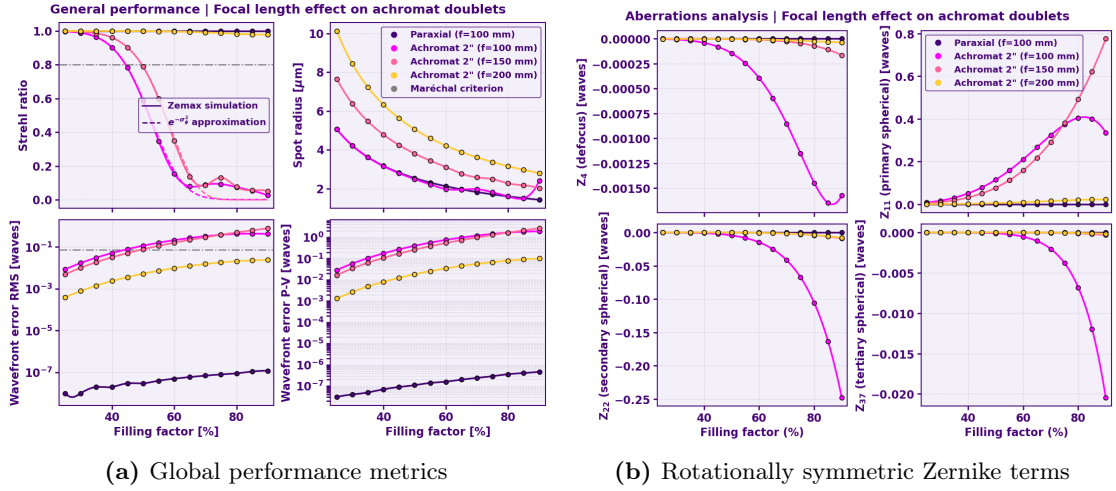


Figure 2.4: Performance evolution in 2" achromatic doublets as a function of focal length. Increasing the focal distance drastically improves wavefront flatness; the $f = 200$ mm variant suppresses Zernike aberrations to near-ideal levels. This improvement implies a spatial trade-off, as the focal spot radius grows linearly with focal length ($w_0 \propto f$), requiring a balance between phase quality and desired power density.

For achromatic doublets (Fig. 2.4), increasing the focal length substantially improves phase flatness. The $f = 200$ mm variant suppresses all evaluated Zernike orders to near-zero levels, maintaining diffraction-limited quality ($S \geq 0.8$) across the full range of pupil filling factors and approaching custom aspheric performance. While the Zernike curves for the $f = 100$ mm doublet appear to rise sharply, inspecting the absolute vertical axis confirms these values remain relatively moderate, appearing exaggerated mainly due to the near-perfect phase suppression of the 150 mm and 200 mm doublets. However, this aberration correction incurs a spatial trade-off, as the focal spot radius scales linearly with focal length ($w_0 \propto f$).

Crucially, this result highlights a highly efficient, cost-effective system architecture for

single-atom detection: long focal length doublets ($f \geq 200$ mm) can be deployed in relay stages where high phase fidelity is required over long transport paths without demanding extreme spatial concentration, *reserving a single high-performance aspheric lens for the final tube stage to achieve optimal focusing* and maximum fluorescence collection.

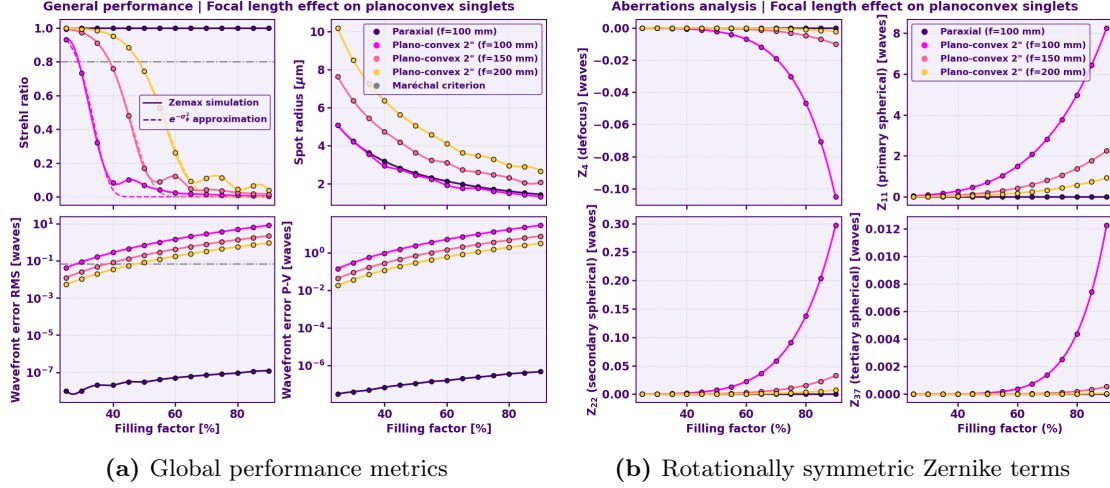


Figure 2.5: Aberration dynamics in 2'' plano-convex singlets. Similar to doublets, longer focal lengths mitigate phase error, but singlets exhibit significantly higher magnitudes of spherical aberration and coma (Z_{11}, Z_{22}, Z_{37}) than their achromatic counterparts. This confirms that, for a given filling factor, the singlet is the least robust option for high-aperture fluorescence collection.

An analogous trend is observed for plano-convex singlets in Fig. 2.5, where longer focal lengths systematically reduce phase errors. Notably, the $f = 200$ mm singlet exhibits Strehl degradation and Zernike aberration magnitudes comparable to those of the $f = 100$ mm doublet, though the latter still retains superior primary spherical aberration (Z_{11}) correction. As with doublets, suppressing wavefront error via longer focal lengths comes at the expense of focal spot concentration.

2.4.4 Diameter effect

To evaluate the influence of physical lens aperture and numerical aperture limits, Fig. 2.6 compares four achromatic doublets structured into two distinct matched focal ratio ($f/\#$, or F -number) pairs: an $f/2$ group consisting of 1" ($f = 50$ mm) and 2" ($f = 100$ mm) variants, and an $f/4$ group pairing 1" ($f = 100$ mm) and 2" ($f = 200$ mm) lenses. The F -number is intrinsically linked to the numerical aperture (NA) in air ($n \simeq 1$) through the relation $NA \simeq \frac{1}{2(f/\#)}$, explicitly quantifying the geometric light-gathering capacity and marginal ray acceptance angles of each optical configuration.

As shown in Fig. 2.6, global optical performance groups predominantly by focal ratio rather than absolute physical aperture. Within each $f/\#$ family, lenses exhibit nearly identical Strehl ratio drops and Zernike aberration scaling, though the 1" diameter variants maintain slightly superior phase flatness due to reduced bulk glass path and edge thickness constraints. However, the performance gap between $f/2$ and $f/4$ systems remains far more dominant, confirming that operating at higher $f/\#$ values is the most effective strategy for suppressing primary and higher-order spherical aberrations.

2.5 Selected configurations and comparative evaluation

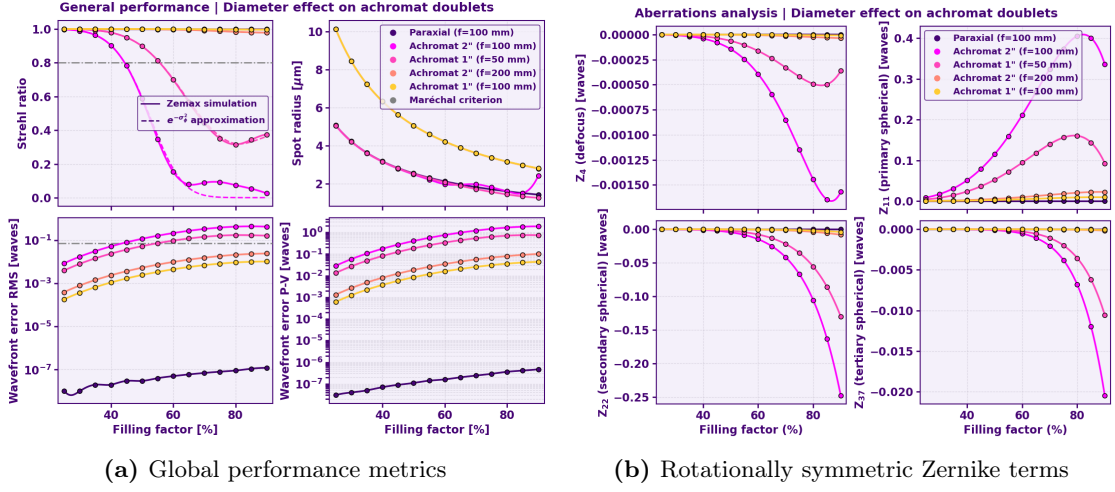


Figure 2.6: Influence of physical diameter and focal ratio ($f/\#$) on performance. Results demonstrate that optical quality is dominated by the focal ratio; $f/4$ systems show superior aberration suppression compared to $f/2$ setups. Notably, 1'' diameters maintain slightly higher phase flatness than 2'' variants due to reduced bulk glass thickness effects, although they offer less mechanical flexibility.

These findings yield a clear design guideline for fluorescence collection setups: *large-diameter, long-focal-length elements* (high $f/\#$) should be prioritized in relay stages to mitigate wavefront degradation and preserve phase flatness across long transport paths. However, this choice involves explicit design trade-offs: longer focal lengths inherently broaden the focused spot (due to intrinsic diffraction scaling $w_0 \propto f$), reducing peak optical power density at the focal plane and increasing potential spatial crosstalk if used directly at the camera interface. Consequently, an optimal setup balances these mechanisms by utilizing long-focal-length doublets for aberration-free relay propagation, while reserving a dedicated high-NA aspheric lens for the final tube stage to achieve tight spatial focusing onto the detector.

2.5 Selected configurations and comparative evaluation

Building upon the single-lens parametric studies and the design constraints established in Section 2.2.3, an automated optimization loop was implemented to systematically evaluate lens combinations from the commercial Thorlabs catalog. To ensure optimal transmission and minimize parasitic reflections at the optical trapping wavelength ($\lambda = 780$ nm), the search space was restricted to standard B-coated optical elements.

To prevent severe wavefront degradation while preserving computational flexibility and operational alignment tolerances, the pupil filling factor (FF) thresholds derived in the previous section were enforced across all candidate systems:

$$FF_{\text{singlets}} \leq 0.30, \quad FF_{\text{achromats}} \leq 0.75, \quad FF_{\text{aspherics}} \leq 0.90. \quad (2.17)$$

The constraint on doublets is set to 0.75; while long focal length achromats with favorable F -numbers can operate at larger pupil fractions without significant phase distortion, this threshold serves as a robust baseline across arbitrary catalog combinations. Furthermore,

capping the maximum pupil filling across all geometries ensures a clear spatial margin around the clear aperture, accommodating potential off-axis beam translation and slight transverse misalignment within the relay telescope stages without inducing aperture clipping.

The field of view for the ray-tracing evaluation was set to encompass the full effective numerical aperture of the optical trapping system ($\text{NA}_{\text{eff}} = 0.65$), corresponding to an object-plane spatial extent of 0.65 mm. At the time of system modeling, the final collection objective had not been physically acquired; hence, an intermediate focal length of $f_{\text{obj}} = 17.5$ mm (representing the expected manufacturer range between 15 mm and 20 mm) was adopted as the reference baseline. Adopting this intermediate FoV rather than the maximum achievable physical area acts as a pragmatic baseline during preliminary screening; it significantly expands the viable catalog search space, allowing a broader diversity of commercial lens pairings to satisfy physical packaging constraints while providing a consistent comparative metric for off-axis performance. Under these optical parameters, the geometric limits for system magnification were constrained between an upper bound of $M_{\text{max}} = 16.40\times$ for complete FoV enclosure and a lower bound of $M_{\text{min}} = 0.46\times$ to guarantee individual pixel isolation on the detector plane.

Executing the search loop across these physical constraints yielded a ranked list of viable multi-lens architectures. From this parameter space, representative configurations spanning distinct optical trade-offs (balancing spatial compactness, total magnification, and wavefront error suppression) were down-selected for detailed end-to-end ray-tracing and physical optics analysis in Zemax OpticStudio.

2.5.1 Optimization loop results and candidate selection

Executing the automated search script revealed stark contrasts in overall design viability among optical families. Standard plano-convex singlets failed completely (0 valid candidates out of 5,832 evaluated states), driven primarily by excessive magnification (60.3% rejected for exceeding M_{max}) and track length violations (25.0%). This confirms that simple singlets cannot satisfy high-NA fluorescence collection constraints under strict pupil filling limits. Conversely, achromatic doublets yielded 7 viable multi-lens combinations, limited predominantly by the upper magnification bound (50.9%) and commercial aperture availability (29.9%). Aspheric lenses provided the highest design versatility with 23 valid configurations, albeit incurring a significantly higher component cost ($3\times$ to $6\times$ that of doublet systems).

To systematically evaluate trade-offs between target magnification mapping and wavefront error suppression, six representative optical systems were down-selected for end-to-end physical optics simulation in Zemax OpticStudio, as summarized in Table 2.1. While these six baseline setups prioritize homogeneous lens families to establish clear comparative bounds, combining achromatic doublets in the relay stages with an aspheric tube lens presents a compelling hybrid architecture. This combination leverages the phase preservation of long-focal achromats during beam transport alongside the tight focusing capability of aspherics at the sensor plane, forming a prime candidate for future experimental implementation.

The selection pairs systems to isolate key physical mechanisms. ACH-1 ($M_{\text{tot}} = 5.71\times$) and

2.5 Selected configurations and comparative evaluation

ID	f_{r1} (mm)	f_{r2} (mm)	f_{tube} (mm)	M_{total}	Track length (cm)	Cost (€)
<i>Achromat doublets</i>						
ACH-1	100	100	100	$5.71\times$	51.75	578.85
ACH-2	100	100	150	$8.57\times$	56.75	578.85
ACH-3	150	100	100	$8.57\times$	61.75	514.16
<i>Aspheric lenses</i>						
ASP-1	100	200	150	$4.29\times$	76.75	3099.28
ASP-1	200	200	100	$5.71\times$	91.75	3714.38
ASP-3	100	200	200	$5.71\times$	81.75	3714.38

Table 2.1: Down-selected commercial multi-lens architectures evaluated in Zemax OpticStudio.

ASP-1 ($M_{\text{tot}} = 4.29\times$) provide the closest commercial match to the target magnification for mapping optical trap spacing onto the camera pixels. To test at a fixed magnification whether a symmetric relay ($M_{\text{relay}} = 1\times$) mitigates aberrations—exploiting how structural symmetry naturally cancels lower-order phase errors—matched pairs were defined within each family: ACH-2 ($1\times$ relay) versus ACH-3 ($0.67\times$ relay) at $M_{\text{tot}} = 8.57\times$, and ASP-2 ($1.00\times$ relay) versus ASP-3 ($2.00\times$ relay) at $M_{\text{tot}} = 5.71\times$.

2.5.2 Performance evaluation of near target magnification configurations

To evaluate the optical systems closest to the target magnification required for camera pixel mapping, the achromatic doublet pair ACH-1 ($M_{\text{tot}} = 5.71\times$) and the aspheric pair ASP-1 ($M_{\text{tot}} = 4.29\times$) were analyzed in detail. The paraxial ray-tracing schematics for both configurations are illustrated in Fig. 2.7. These diagrams provide a visual reference for the physical disposition, relative element spacings, and longitudinal track lengths of each candidate system. It must be noted that these paraxial sketches serve exclusively to illustrate spatial layouts; exact optical performance evaluation relies on full wave-optical simulations conducted in Zemax OpticStudio at $\lambda = 780$ nm. The simulations evaluated light emitted from both the central (0.00 mm) and extreme (0.325 mm) FoV positions, fully filling the numerical aperture ($\text{NA} = 0.65$) of the trapping collection system.

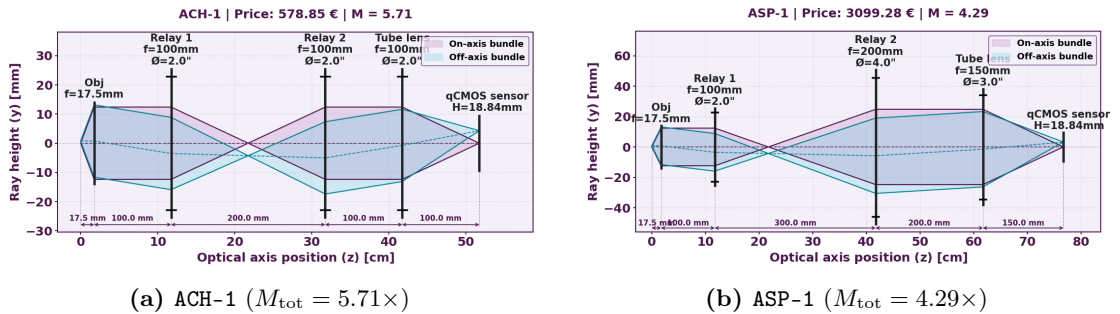


Figure 2.7: Paraxial ray-tracing schematics for (a) ACH-1 and (b) ASP-1. These visual diagrams outline element disposition and axial spacing scales. Quantitative optical performance is derived from full Zemax OpticStudio simulations at $\lambda = 780$ nm.

The resulting diffraction and beam spatial distribution metrics are summarized in Fig. 2.8. Across the entire FoV, the aspheric architecture (ASP-1) outperforms the achromatic doublet design (ACH-1). However, evaluating the Strehl ratio as a consolidated figure of merit reveals that both configurations fall well below Maréchal’s diffraction limit ($S \geq 0.80$), with ACH-1 exhibiting lower overall phase quality ($S < 0.25$) compared to ASP-1 ($S \simeq 0.35$). This wavefront degradation causes the point spread function (PSF) of the achromatic system to smear considerably; for typical scientific camera pixel pitches ($\sim 5\text{--}13\ \mu\text{m}$), *the enlarged aberrated spot footprint extends energy across an area equivalent to 3–4 resolution elements rather than remaining confined within 1–2 pixels*, directly degrading signal-to-background contrast. Furthermore, off-axis evaluation at the extreme FoV reveals an asymmetric spot profile ($\text{Spot}_x \neq \text{Spot}_y$) in both configurations, driven predominantly by field-dependent astigmatism and off-axis coma. While ASP-1 also suffers from this off-axis axial disparity, its overall spot radius remains vastly more compact, preserving pixel-limited optical isolation between adjacent trapping sites.

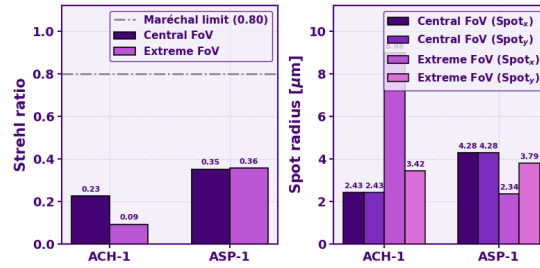


Figure 2.8: Performance metrics for ACH-1 and ASP-1 systems at the center and extreme of the field. The aspheric design (ASP-1) guarantees a compact spot and superior Strehl ratio across the entire FoV. Conversely, the achromatic system (ACH-1) suffers from severe off-axis PSF blurring ($\text{Spot}_x \neq \text{Spot}_y$), which would significantly increase crosstalk between adjacent trapping sites.

2.5.3 Effect of relay symmetry on optical performance

To evaluate whether structural symmetry in the relay stage ($M_{\text{relay}} = 1.00\times$) mitigates off-axis aberrations, matched system pairs were analyzed within each lens family: ACH-2 ($1.00\times$ relay) versus ACH-3 ($0.67\times$ relay) at $M_{\text{tot}} = 8.57\times$, and ASP-2 ($1.00\times$ relay) versus ASP-3 ($2.00\times$ relay) at $M_{\text{tot}} = 5.71\times$. The Strehl ratio and spot radius metrics for these four systems are presented in Fig. 2.9.

The macro-level metrics reveal a counterintuitive trend regarding relay symmetry across optic families. For achromatic doublets, enforcing relay symmetry (ACH-2) degrades system performance compared to the asymmetric design (ACH-3). Conversely, in aspheric architectures, relay symmetry (ASP-2) improves optical quality over the asymmetric alternative (ASP-3), *making the symmetric ASP-2 layout the highest-performing aspheric system overall*. Regarding spot dimensions, achromatic configurations again display an exorbitant, uncontrolled growth of the spot radius along one axis at the extreme field edge. In contrast, aspheric designs keep the spot size tightly bounded across the entire field, though a slight disparity between orthogonal axes ($\text{Spot}_x \neq \text{Spot}_y$) emerges off-axis.

To uncover the physical mechanisms behind these behaviors, a wavefront decomposition

2.5 Selected configurations and comparative evaluation

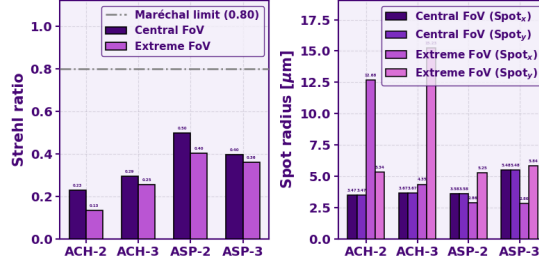


Figure 2.9: Optical performance metrics for symmetric and asymmetric relay configurations (ACH-2/3 and ASP-2/3) evaluated at central (0.00 mm) and extreme (0.325 mm) FoV positions. Left: Strehl ratio relative to the Maréchal limit ($S = 0.80$). Right: Calculated spot radii along orthogonal axes (Spot_x and Spot_y).

into Zernike polynomials was conducted, as detailed in Sec. 2.3.4. The primary Zernike coefficients (Z_4 through Z_{11}) for both field positions are shown in Fig. 2.10.

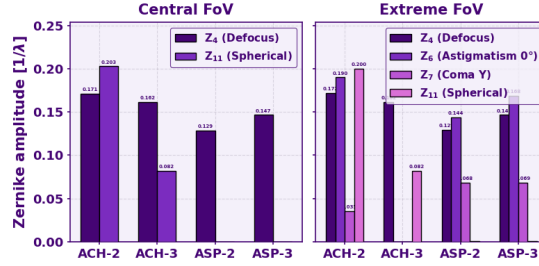


Figure 2.10: Zernike expansion coefficients (Z_4 to Z_{11}) for central and extreme FoV positions across relay configurations, illustrating the dominant aberration modes.

The modal analysis demonstrates that achromatic doublets are heavily bottlenecked by primary spherical aberration (Z_{11}), which remains severe across both central and off-axis fields regardless of relay symmetry. Aspheric surfaces successfully eliminate spherical aberration entirely; however, off-axis light propagation through the aspheric relay introduces field-dependent coma (Z_7, Z_8) and astigmatism (Z_5, Z_6). These off-axis modes break the rotational symmetry of the central spot profile, accounting for the axial spot anisotropy observed in the extreme field of view.

In summary, among the evaluated commercial candidates, symmetric aspheric configurations (such as ASP-2) yield the highest overall performance and come closest to diffraction-limited collection across the target field of view. While achieving high energy concentration is critical for minimizing optical cross-talk between closely spaced traps prior to sensor sampling, *these findings strongly point toward a hybrid design as the optimal practical compromise*: leveraging cost-effective long-focal-length achromatic doublets in the relay stage to preserve phase during transport, paired with a dedicated aspheric tube lens to maximize final spot compactness.

3 Detection stage: signal estimation and noise analysis

This chapter details the theoretical and physical framework governing the optical detection pipeline. The analysis begins with an estimation of the expected atomic fluorescence, quantifying the photon flux emitted by the atoms and determining the exact fraction collected by the imaging optics onto the camera sensor. Subsequently, the operational mechanisms and dominant noise sources of various photodetector architectures—specifically Charge-Coupled Devices (CCD), Electron-Multiplying CCDs (EMCCD), and quantitative Complementary Metal-Oxide-Semiconductor (qCMOS) cameras—are examined. By integrating these optical losses and detector noise profiles, analytical expressions for the Signal-to-Noise Ratio (SNR) are derived for each technology, establishing a quantitative baseline to evaluate performance trends under nominal photon flux conditions.

3.1 Expected atomic fluorescence and photon collection

To establish a rigorous and realistic signal baseline, we model the total photon flux emitted by a single ^{87}Rb atom localized within an optical tweezer and quantify the fraction collected by the imaging optics onto the detector plane.

3.1.1 Single-atom scattering rate

When a single atom is illuminated by near-resonant light, its optical response is fundamentally governed by its natural linewidth Γ , which represents the natural spontaneous emission rate from the excited state. For the D_2 transition of ^{87}Rb ($5^2\text{S}_{1/2} \rightarrow 5^2\text{P}_{3/2}$), this parameter is $\Gamma \simeq 2\pi \times 6.065 \text{ MHz}$ [58]. In an ideal free-space two-level system under strong driving intensity, the scattering rate asymptotically approaches the theoretical maximum of $\Gamma/2$. However, in a tightly focused optical tweezer, the effective photon emission rate is significantly lower due to the light-shift induced by the trapping field and the operating regime of the cooling beams.

As stated in Chap. 2, the beam waist ω_0 and potential depth U_0 scale directly with the effective numerical aperture NA_{eff} of the trapping line (see Eq. (2.5)). While increasing NA_{eff} tightens the spatial confinement, the depth of the potential well U_0 inherently corresponds to the AC Stark shift experienced by the atomic ground state ($\hbar\Delta_{\text{shift},g} = -U_0$). Because the excited state exhibits a different polarizability, this local field introduces a differential AC Stark shift ($\Delta_{\text{diff}} = \Delta_{\text{shift},e} - \Delta_{\text{shift},g}$) between the

3.1 Expected atomic fluorescence and photon collection

ground and excited levels. This differential shift displaces the D_2 atomic transition relative to the Magneto-Optical Trap (MOT) cooling beams, resulting in an effective detuning $\Delta_{\text{eff}} = \Delta_{\text{laser}} - \Delta_{\text{diff}}$, where Δ_{laser} represents the nominal red-detuning of the cooling lasers.

This trap-induced detuning shifts the atom off-resonance, suppressing the excitation efficiency driven by the cooling beams. In the single-atom loading regime, this off-resonance condition is parameterized by an effective saturation parameter s_{eff} :

$$s_{\text{eff}} = \frac{I/I_{\text{sat}}}{1 + 4 \left(\frac{\Delta_{\text{eff}}}{\Gamma} \right)^2}, \quad (3.1)$$

where I/I_{sat} is the normalized intensity of the cooling beams (typically assuming the atom experiences the intensity of one effective cooling beam on average). The resulting photon scattering rate R_{scat} scales according to:

$$R_{\text{scat}} = \frac{\Gamma}{2} \frac{s_{\text{eff}}}{1 + s_{\text{eff}}}. \quad (3.2)$$

Qualitatively, this introduces an operational trade-off that is actively managed in single-atom experiments. In practice, a high- NA_{eff} regime is deliberately chosen to form sub-micron focal spots with deep potential wells, which are essential for reaching the single-atom collisional blockade regime. Although the resulting high optical intensity drives a strong ground-state AC Stark shift that initially pushes the transition far off-resonance ($\Delta_{\text{eff}} \gg \Gamma$) and suppresses R_{scat} , this shift is routinely compensated during the readout phase. Rather than reducing the numerical aperture, experimental protocols dynamically adjust the cooling laser detuning (Δ_{laser}) during imaging to offset Δ_{diff} , restoring the effective saturation parameter s_{eff} and maximizing the photon scattering rate.

To establish a realistic order-of-magnitude estimate for our photon budget without loss of generality, we adopt representative operational parameters validated in typical single-atom optical tweezer setups [67] (e.g., $\text{NA}_{\text{eff}} = 0.5$, dipole power $P \simeq 4$ mW). Under these conditions, the trap-induced light-shift reaches $\Delta_{\text{shift}} \simeq 2\pi \times 25$ MHz. Combined with a typical cooling laser detuning $\Delta_{\text{laser}} \simeq -2\pi \times 30$ MHz, the atom experiences an effective detuning of $\Delta_{\text{eff}} \simeq -2\pi \times 55$ MHz. This yields an effective saturation parameter of $s_{\text{eff}} \simeq 0.05$ and a baseline scattering rate of:

$$R_{\text{scat}} \simeq 1.0 \times 10^6 \text{ photons/s}. \quad (3.3)$$

This benchmark scattering rate provides the foundation for estimating the photon flux collected at the sensor plane.

3.1.2 Geometric collection and optical losses

The atomic fluorescence is emitted isotropically over a full solid angle of 4π steradians. The collection objective acts as a spatial aperture that captures a cone of light characterized by its numerical aperture NA_{coll} and acceptance half-angle $\theta = \arcsin(\text{NA}_{\text{coll}})$. Crucially, while the trapping aperture NA_{eff} dictates the local trap depth and the atomic scattering

rate R_{scat} , the collection objective itself remains fixed at $\text{NA}_{\text{coll}} = 0.65$. Thus, the collection geometry is invariant across different trapping configurations.

To determine the solid angle Ω captured by the collection optics, we integrate the differential solid angle element $d\Omega = \sin \theta' d\theta' d\phi$ over the cone of acceptance from the optical axis out to the marginal ray boundary:

$$\Omega = \int_0^{2\pi} d\phi \int_0^\theta \sin \theta' d\theta' = 2\pi (1 - \cos \theta). \quad (3.4)$$

Substituting the identity $\cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \text{NA}_{\text{coll}}^2}$ yields the explicit solid angle expression:

$$\Omega = 2\pi \left(1 - \sqrt{1 - \text{NA}_{\text{coll}}^2} \right). \quad (3.5)$$

Normalizing this captured solid angle by 4π steradians defines the fixed geometric collection efficiency of the objective lens:

$$\eta_{\text{geom}} = \frac{\Omega}{4\pi} = \frac{1 - \sqrt{1 - \text{NA}_{\text{coll}}^2}}{2}. \quad (3.6)$$

For our constant collection aperture of $\text{NA}_{\text{coll}} = 0.65$, this evaluates to $\eta_{\text{geom}} \simeq 0.1201$, meaning that the objective captures 12.01% of the isotropically emitted photons.

Beyond geometric acceptance, photons traveling through the downstream optical line at $\lambda_{\text{flu}} = 780$ nm suffer from deterministic reflection and absorption losses. Because the final collection objective was not physically acquired at the time of system modeling, internal transmission losses were modeled using a representative baseline of $\eta_{\text{obj}} \simeq 0.87$, drawn from established experimental literature for single-atom collection systems [67]. A dichroic mirror used to separate the 850 nm trap light from the 780 nm fluorescence contributes an efficiency of $\eta_{\text{dichroic}} \simeq 0.90$, while a narrow band-pass filter used to reject stray ambient background provides $\eta_{\text{filter}} \simeq 0.90$. Finally, the relay optics and tube lens introduce $N = 6$ air-glass interfaces; with broadband anti-reflective coatings ($R \leq 0.01$), the multi-surface Fresnel transmission is $(1 - R)^N \simeq 0.9415$. Multiplying these contributions gives a net cumulative optical transmission efficiency of $\eta_{\text{trans}} = \eta_{\text{obj}} \cdot \eta_{\text{dichroic}} \cdot \eta_{\text{filter}} \cdot (1 - R)^N \simeq 0.6635$.

Combining the constant geometric acceptance with the net transmission factor defines the overall system efficiency $\eta_{\text{tot}} = \eta_{\text{geom}} \cdot \eta_{\text{trans}} \simeq 0.1201 \times 0.6635 \simeq 0.0797$ (7.97%). The expected number of photons incident on the detector array N_{inc} over an exposure time t_{exp} is given by:

$$N_{\text{inc}} = \eta_{\text{tot}} \cdot R_{\text{scat}} \cdot t_{\text{exp}}. \quad (3.7)$$

Using the baseline scattering rate $R_{\text{scat}} \simeq 1.0 \times 10^6$ photons/s and a standard integration time of $t_{\text{exp}} = 10$ ms, the atom emits approximately $N_{\text{emitted}} \simeq 1.0 \cdot 10^4$ photons. Accounting for the overall optical efficiency $\eta_{\text{tot}} \simeq 0.0797$, the expected number of photons incident on the sensor plane is $N_{\text{inc}} \simeq 797$ photons per 10 ms exposure. It is important to emphasize that *these values do not represent the definitive metrics of our specific experimental realization*, but rather serve as a reference to establish the relevant order of magnitude.

3.2 Photodetector architectures and noise mechanisms

The optical detection of a single trapped atom via fluorescence imaging is fundamentally formulated as a binary hypothesis testing problem. We define two competing operational states: the null hypothesis H_0 , representing an empty optical tweezer where the registered signal consists purely of sensor-induced noise and stray laser scattering, and the alternative hypothesis H_1 , representing the presence of a single ^{87}Rb atom contributing an additional fluorescence photon flux.

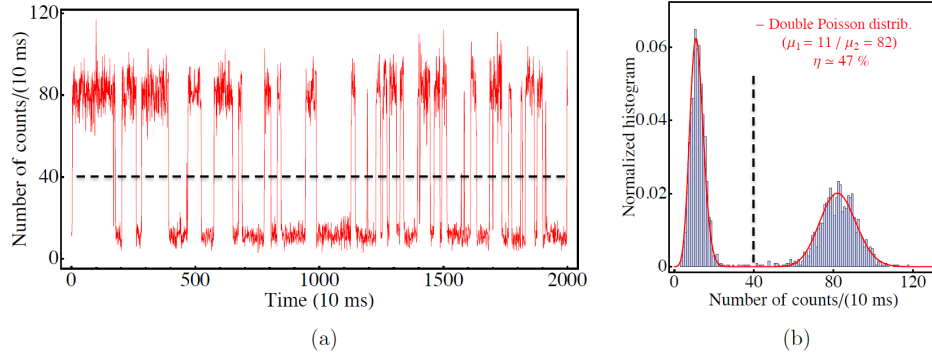


Figure 3.1: Experimental signature of single-atom trapping [67]. (a) Time trace of fluorescence counts per 10 ms bin showing discrete quantum jumps between H_0 (empty trap) and H_1 (single atom present). (b) Corresponding normalized count histogram revealing a clear bimodal distribution fitted by a double Poissonian function with means $\mu_1 = 11$ and $\mu_2 = 82$ counts per 10 ms bin.

In experimental setups, recording time-resolved loading traces yields discrete quantum jumps between these two regimes, as shown in Fig. 3.1(a). Aggregating thousands of consecutive integration intervals results in a characteristic bimodal histogram of digital counts (Fig. 3.1(b)). The left Poissonian peak corresponds to H_0 , centered at a baseline background level μ_0 with variance σ_0^2 , while the right peak corresponds to H_1 , centered at a higher signal mean $\mu_1 = \mu_0 + \Delta\mu$ with variance σ_1^2 . The separation between these peaks determines the fidelity of single-atom state discrimination.

The fidelity of the atomic state readout depends strictly on the physical separation between these two distributions relative to their respective widths. A significant overlap between H_0 and H_1 introduces classification errors, where thermal or shot noise fluctuations in an empty trap trigger false positives, or low fluorescence counts cause false negatives. While a complete evaluation of readout fidelity requires integration over the overlapping distribution tails—which can be complicated by non-Gaussian multiplication statistics or sparse photon counting—the Signal-to-Noise Ratio (SNR) provides a practical first-order scalar metric. By quantifying the ratio of signal separation to average distribution width, the SNR serves as a reliable proxy for state discriminability under near-Gaussian assumptions.

$$\text{SNR} = \frac{S_{\text{net}}}{\sigma_{\text{total}}}. \quad (3.8)$$

Here, $S_{\text{net}} = \mu_1 - \mu_0$ isolates the net mean atomic signal $\Delta\mu$ after background subtraction protocols. For independent measurements of the empty trap (H_0) and the atom-loaded trap (H_1), the noise variances combine in quadrature during subtraction, such that the

total variance of the differential signal is strictly given by $\sigma_{\text{total}}^2 = \sigma_0^2 + \sigma_1^2$. Maximizing this SNR compresses these statistical widths relative to their mutual separation $\Delta\mu$, pushing the bimodal distribution toward complete statistical isolation. Evaluating this metric across sensor architectures establishes a closed-form analytical framework to identify optimal hardware configurations, reserving full histogram overlap integration and threshold optimization as further research.

Because the underlying physical noise processes—atomic photon shot noise (σ_{shot}), background photon shot noise (σ_{bg}), dark current thermal excitations (σ_{dark}), and electronic readout noise (σ_{read})—stem from uncorrelated microscopic and electronic events, their mutual covariances vanish. Consequently, the variance of their sum equals the sum of their individual variances, allowing the total noise envelope to be evaluated in quadrature as $\sigma_{\text{total}}^2 = \sigma_{\text{shot}}^2 + \sigma_{\text{bg}}^2 + \sigma_{\text{dark}}^2 + \sigma_{\text{read}}^2$. Environmental background scattering and thermal dark current scale as fractional components of the incident photon flux through empirical parameters $f_{\text{bg}} \simeq 10^{-3}$ and $f_{\text{dark}} \simeq 10^{-8}$ [67].

3.2.1 Charge-Coupled Devices (CCDs)

The conventional Charge-Coupled Device (CCD) operates by converting incident photons into photo-electrons within a continuous matrix of silicon potential wells. The defining feature of a standard CCD is its sequential readout mechanism. At the end of the exposure window, parallel vertical clock voltages shift the stored charge packets row-by-row into a horizontal shift register. *These charges are then transferred serially, pixel-by-pixel, toward a single output node located at the corner of the chip, where charge-to-voltage conversion takes place.* While reading the entire image through a single shared amplifier guarantees high spatial uniformity, it imposes a strict bandwidth limit: operating at higher readout rates to increase frame speed inherently degrades the charge conversion accuracy and elevates the electronic read noise floor σ_{read} .

To derive the governing SNR, we evaluate the signal and noise contributions integrated over a binned super-pixel array of $N_{\text{bin}} \times N_{\text{bin}}$ pixels. The net atomic signal S_{CCD} collected with a sensor quantum efficiency η_q and its corresponding total noise variance σ_{CCD}^2 are formulated as:

$$\begin{aligned} S_{\text{CCD}} &= \eta_q \cdot N_{\text{photons}}, \\ \sigma_{\text{CCD}}^2 &= \eta_q N_{\text{photons}} (1 + f_{\text{bg}} N_{\text{bin}}^2) + N_{\text{photons}} f_{\text{dark}} N_{\text{bin}}^2 + \sigma_{\text{read}}^2. \end{aligned} \tag{3.9}$$

Because charge packets are shifted and aggregated in the analog domain within the physical registers prior to output, the read noise variance σ_{read}^2 is added only once per readout cycle, giving $\text{SNR}_{\text{CCD}} = S_{\text{CCD}}/\sigma_{\text{CCD}}$. For full algebraic derivations and parameter definitions of each individual term in Eq. (3.9), the interested reader is referred to the interactive Python notebooks provided in the project repository [66].

3.2.2 Electron-Multiplied CCDs (EMCCDs)

The Electron-Multiplied CCD (EMCCD) was developed specifically to bypass the electronic read noise floor that restricts conventional CCDs in low-light applications. The physical architecture mirrors a standard CCD but *integrates an extended solid-state*

multiplication register between the horizontal shift register and the final output video amplifier. As photo-electrons pass through the stages of this multiplication register, they are driven by high clock voltages that accelerate them to kinetic energies sufficient to trigger impact ionization in the silicon lattice. This generates a cascade of secondary electrons, amplifying the charge packet by a user-defined multiplication gain G prior to digitization. This pre-amplification effectively reduces the output read noise in the signal domain to σ_{read}/G , enabling single-photon detection when $G \gg \sigma_{\text{read}}$.

However, impact ionization is a stochastic process governed by probabilistic secondary emission at each stage. This introduces statistical fluctuations into the gain mechanism, quantified by an excess noise factor $F \simeq \sqrt{2}$ at high gain levels. This amplification uncertainty effectively doubles the variance of all photon shot noise and pre-amplified noise channels entering the register. Accounting for Clock-Induced Charge (σ_{CIC}) generated by high-voltage clock transitions, the amplified signal S_{EMCCD} and its associated total noise variance σ_{EMCCD}^2 are given by:

$$\begin{aligned} S_{\text{EMCCD}} &= G \cdot \eta_q \cdot N_{\text{photons}}, \\ \sigma_{\text{EMCCD}}^2 &= G^2 F^2 (\eta_q N_{\text{photons}} (1 + f_{\text{bg}} N_{\text{bin}}^2) + N_{\text{photons}} f_{\text{dark}} N_{\text{bin}}^2 + \sigma_{\text{CIC}} N_{\text{bin}}^2) + \sigma_{\text{read}}^2. \end{aligned} \quad (3.10)$$

Taking the ratio yields $\text{SNR}_{\text{EMCCD}} = S_{\text{EMCCD}}/\sigma_{\text{EMCCD}}$. The factor $F^2 \simeq 2$ imposes a fundamental penalty on the effective quantum efficiency, meaning EMCCDs require higher photon numbers to achieve shot-noise-limited performance compared to an ideal noise-free detector.

3.2.3 Quantitative Complementary Metal-Oxide-Semiconductor (qCMOS)

Scientific Complementary Metal-Oxide-Semiconductor (sCMOS) sensors eliminate the sequential readout bottleneck of CCDs by distributing charge-to-voltage conversion across the entire matrix, *equipping every individual pixel column with its own dedicated amplifier and analog-to-digital converter.* Quantitative CMOS (qCMOS) technology refines this architecture by drastically reducing the physical capacitance C of the pixel floating diffusion node. Since the conversion gain scales as $\Delta V = \Delta Q/C$, suppressing parasitic capacitance maximizes the voltage response generated per photo-electron, pushing the signal far above the thermal noise floor of the readout transistors.

This passive electronic suppression achieves a sub-electron read noise floor ($\sigma_{\text{read}} \sim 0.2\text{--}0.3 e^-$ RMS) without requiring active multiplication. Consequently, the excess noise factor remains strictly unity ($F = 1$), avoiding the noise doubling penalty characteristic of EMCCDs. Because digitization occurs locally at each pixel prior to spatial aggregation, software-level pixel binning over an $N_{\text{bin}} \times N_{\text{bin}}$ cluster causes the independent read noise variances of individual pixels to sum in quadrature. The signal S_{qCMOS} and noise variance σ_{qCMOS}^2 are expressed as:

$$\begin{aligned} S_{\text{qCMOS}} &= \eta_q \cdot N_{\text{photons}}, \\ \sigma_{\text{qCMOS}}^2 &= \eta_q N_{\text{photons}} (1 + f_{\text{bg}} N_{\text{bin}}^2) + N_{\text{photons}} f_{\text{dark}} N_{\text{bin}}^2 + \sigma_{\text{read}}^2 N_{\text{bin}}^2. \end{aligned} \quad (3.11)$$

The resulting Signal-to-Noise Ratio is $\text{SNR}_{\text{qCMOS}} = S_{\text{qCMOS}}/\sigma_{\text{qCMOS}}$. Combining high quantum efficiency η_q with sub-electron read noise and an excess noise factor of $F = 1$

ensures that qCMOS architectures maintain a shot-noise-limited response down to low-photon flux regimes, making them optimal for fast, high-SNR fluorescence imaging.

3.2.4 Fundamental quantum noise limits

The performance of all three detector architectures transitions through distinct operational regimes depending on the incident photon flux N_{photons} . At ultra-low photon levels, electronic noise sources—primarily σ_{read} in conventional CCDs and qCMOS, or σ_{CIC} and residual multiplication noise in EMCCDs—dictate state discriminability. As the collected atomic signal increases, Poissonian photon shot noise eventually dominates over electronic noise floors, *driving system performance toward the Standard Quantum Limit* (SQL), where relative signal precision scales as $1/\sqrt{N_{\text{photons}}}$. However, persistent background scattering f_{bg} prevents practical experimental setups from reaching an ideal zero-background asymptote, maintaining a baseline statistical noise floor regardless of sensor efficiency.

Achieving high readout fidelity within ultra-short exposure times—strictly constrained by atomic heating and loss mechanisms in optical tweezers—requires maximizing the SNR before reaching loss-induced limits. Beyond classical intensity optimization, moving past Poissonian light statistics via non-classical light fields, such as amplitude-squeezed states ($\Delta N^2 < \langle N \rangle$), can suppress photon number fluctuations below the classical shot noise limit. This fundamentally shifts the scaling of state readout fidelity away from the SQL toward the ultimate Heisenberg scaling limit ($1/N$).

3.3 Impact of optical magnification on readout SNR

The overall optical magnification M of the fluorescence imaging system directly dictates the spatial distribution of the atomic point-spread function (PSF) across the sensor active area. Increasing optical magnification expands the spatial footprint of the fluorescence signal, spreading the incident photons over a larger cluster of $N_{\text{bin}} \times N_{\text{bin}}$ pixels (e.g., expanding from an ideal 1×1 focal spot to 2×2 or 3×3 array footprints). While higher magnification reduces optical crosstalk and improves spatial resolution, it distributes the fixed atomic photon yield N_{photons} across multiple detection channels, accumulating independent noise contributions from each involved pixel. *The resulting degradation of the SNR is strongly governed by the underlying sensor readout mechanism and the choice between hardware-level and software-level pixel binning.* In standard CCDs and EMCCDs, native (analog) binning physically shifts and accumulates charge packets within the potential wells prior to charge-to-voltage conversion, contributing the read noise variance σ_{read}^2 only once per readout event. Conversely, post-acquisition digital binning reads every pixel independently and sums their digitized values in software, causing the read noise variance to scale quadratically with the spatial footprint as $N_{\text{bin}}^2 \sigma_{\text{read}}^2$. In contrast, because qCMOS architectures feature localized per-column digitizers, native hardware binning in qCMOS sensors is structurally digital; hence, the operational distinction between native and software binning collapses for qCMOS detectors.

Fig. 3.2 models the evolution of readout SNR as a function of the collected photon flux N_{photons} for representative commercial CCD (Horiba) and EMCCD (Andor) sensors

under varying spatial spot sizes and binning strategies. Both panels benchmark the evaluated performance against the zero-background Standard Quantum Limit (SQL) and the single-pixel baseline (1×1). To isolate the noise penalties induced by optical magnification, spatial spreads of 2×2 and 3×3 pixels are evaluated, comparing the performance gains of native analog charge aggregation against digital post-processing summation.

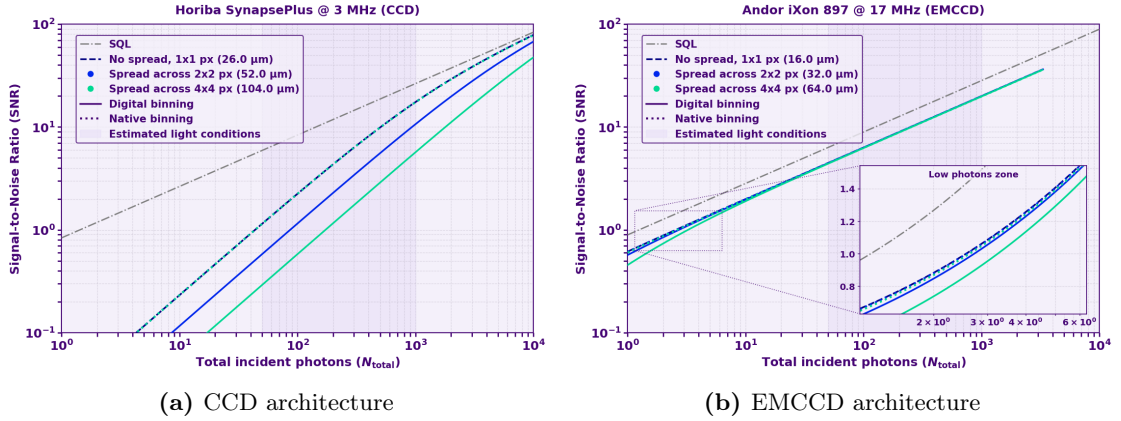


Figure 3.2: Readout Signal-to-Noise Ratio (SNR) scaling versus incident photon flux N_{photons} across different spatial spot sizes and binning strategies for (a) a Horiba CCD and (b) an Andor EMCCD. The response curves highlight performance against the Standard Quantum Limit (SQL), the concentrated 1×1 pixel limit, and extended footprints (2×2 , 3×3) comparing native analog binning against digital software binning.

The numerical results in Fig. 3.2 confirm the crucial role of readout mechanisms when spatial signal spreading occurs. For the conventional CCD (Fig. 3.2a), digital binning introduces severe SNR degradation across extended $N_{\text{bin}} \times N_{\text{bin}}$ footprints due to its elevated baseline read noise floor (σ_{read}). Conversely, native analog binning shifts and aggregates photo-electrons within the physical shift registers prior to readout, effectively recovering the exact SNR performance curve of an unspread 1×1 pixel spot. This theoretical equivalence assumes that the integrated charge within the binned super-pixel remains below the pixel full-well capacity, a condition comfortably met within the photon-sparse regime of single-atom imaging. In the EMCCD architecture (Fig. 3.2b), a similar disparity between digital and native binning is observed at ultra-low photon fluxes ($N_{\text{photons}} \ll 10^2$), where the amplified atomic signal remains comparable to the unamplified output read noise. However, as the photon yield increases, the multiplication gain G effectively eclipses σ_{read} , causing the digital and native binning curves to converge into a single response envelope bounded by the excess noise factor $F \simeq \sqrt{2}$.

To evaluate how sub-electron read noise alters these spatial spreading trade-offs without electron multiplication, we extend this analytical framework to quantitative CMOS technology. Fig. 3.3 illustrates the performance of an Orca qCMOS camera under identical magnification regimes, explicitly quantifying its relative efficiency with respect to the Standard Quantum Limit.

As demonstrated in Fig. 3.3a, the sub-electron read noise characteristic of the Orca qCMOS architecture ($\sigma_{\text{read}} \simeq 0.2\text{--}0.3 \text{ } e^- \text{ RMS}$) induces a behavior analogous to that of

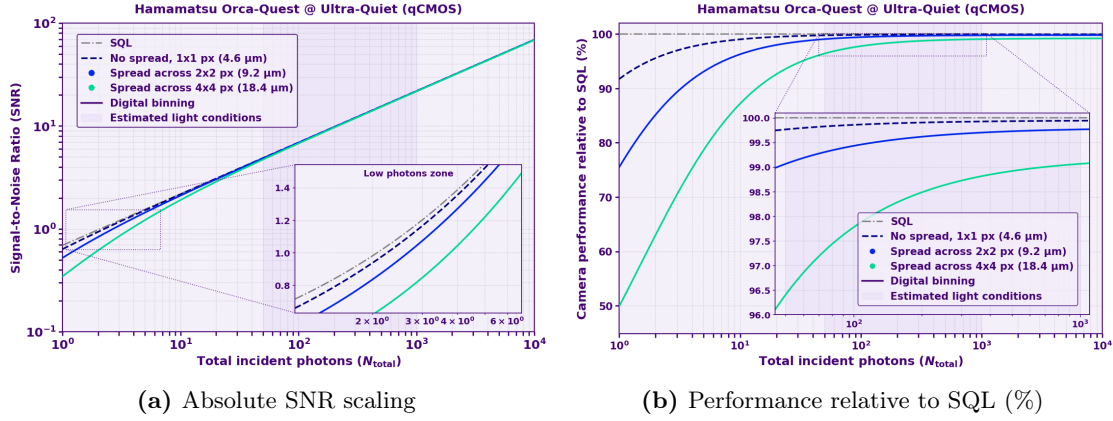


Figure 3.3: Performance analysis of an Orca qCMOS detector under varying optical magnification footprints. (a) Absolute Signal-to-Noise Ratio (SNR) scaling versus photon flux N_{photons} compared against the SQL and a single-pixel limit (1×1). Because qCMOS readout is intrinsically digital, native binning coincides with software aggregation. (b) Fractional performance relative to the theoretical SQL asymptote, highlighting performance separation across spatial footprints (1×1 , 2×2 , 3×3).

the EMCCD, but without the multiplicative noise penalty ($F = 1$). Because the baseline electronic noise floor is remarkably suppressed, even small atomic fluorescence signals quickly eclipse σ_{read} . Consequently, the absolute SNR curves corresponding to different spatial footprints (1×1 , 2×2 , and 3×3) converge rapidly as the photon flux increases. Fig. 3.3b provides a clearer visualization of this separation by evaluating the fractional performance relative to the ideal Standard Quantum Limit. Within the operational photon flux regime typical for single-atom readout ($N_{\text{photons}} \gtrsim 10^2$), the qCMOS sensor consistently operates above 95% of the theoretical SQL, demonstrating high robustness against magnification-induced spatial spreading. While the EMCCD architecture still offers respectable performance by reaching around 70% of the SQL limit, the absence of spurious clock-induced charge and excess noise in qCMOS sensors clearly establishes their superior efficiency for low-light state detection. Conversely, as the incident signal drops below this threshold and becomes comparable to the cumulative multi-pixel read noise floor, the relative efficiency of qCMOS decays rapidly, marking the transition from photon-shot-noise-dominated performance to readout-noise-limited state detection.

In summary, the trade-offs introduced by optical magnification on atomic readout SNR and fidelity depend critically on the underlying sensor architecture and its resilience to multi-pixel signal distribution. It is important to emphasize that optical magnification itself does not alter the total photon yield emitted by the atom or collected by the objective; rather, it distributes the fixed photon flux over a larger number of physical pixels on the sensor grid. While standard CCDs suffer severe performance degradation under spatial spreading due to their high read noise floor—requiring analog binning to mitigate the accumulated electronic noise—both EMCCDs and qCMOS architectures maintain an SNR that is largely insensitive to magnification within the typical operational regime ($N_{\text{photons}} \gtrsim 10^2$). In this domain, the signal remains far above the effective electronic noise floor for both technologies. However, qCMOS achieves superior absolute performance by combining sub-electron read noise with the absence of multiplicative

noise ($F = 1$), operating near the SQL ($> 95\%$), whereas EMCCDs remain intrinsically bounded by the excess noise factor ($F \simeq \sqrt{2}$). Beyond total noise limits, the choice of optimal readout geometry involves balancing additional physical parameters such as sensor frame rates, pixel saturation bounds, and background scatter.

4 Conclusions and further research

4.1 Summary of key findings

This master’s thesis has addressed the critical engineering and physical trade-offs involved in designing a high-efficiency fluorescence collection and imaging pipeline for a neutral-atom quantum platform based on ^{87}Rb Rydberg atom arrays trapped in optical tweezers. The primary objective was to maximize the SNR for state-selective binary readout within ultra-short exposure windows, directly dictating the ultimate state detection fidelity of the quantum processor.

At the optical stage, a systematic parametric investigation of commercial lens architectures—ranging from plano-convex singlets and achromatic doublets to aspheric lenses—demonstrated that standard spherical optics suffer from catastrophic marginal ray spherical aberration (Z_{11}) under high pupil filling factor conditions, degrading focal spot sharpness and reducing single-atom signal collection. While doublets reduce primary aberrations at moderate pupil filling factors, they accumulate significant higher-order spherical terms (Z_{22}, Z_{37}) when fully illuminated, which broadens the spatial spread of the point spread function (PSF) and increases inter-atom spatial crosstalk during multi-atom readout. A grid-search optimization over commercial catalogs identified the symmetric aspheric configuration (**ASP-2**, utilizing a $1 \times 4f$ relay stage) as the optimal solution. Although off-axis light propagation across the FoV introduces field-dependent primary astigmatism (Z_5, Z_6) and coma (Z_7, Z_8) that slightly break spot symmetry ($\text{Spot}_x \neq \text{Spot}_y$), *ASP-2 completely eliminates spherical aberration on-axis and provides superior energy concentration and PSF compactness compared to all doublet alternatives*, ensuring tighter signal enclosure on individual pixels and thereby minimizing state discrimination errors.

At the detection stage, an absolute photon budget was established, modeling the trap-induced AC Stark shift ($\Delta_{\text{shift}} \simeq -2\pi \times 25 \text{ MHz}$) and effective cooling laser detuning ($\Delta_{\text{eff}} \simeq -2\pi \times 55 \text{ MHz}$). With an effective saturation parameter $s_{\text{eff}} \simeq 0.05$ and a scattering rate $R_{\text{scat}} \simeq 1.0 \times 10^6 \text{ photons/s}$, a total system collection efficiency of $\eta_{\text{tot}} = \eta_{\text{geom}} \cdot \eta_{\text{trans}} \simeq 7.97\%$ yields an expected signal of approximately 800 incident photons per 10 ms exposure window. Comparative SNR modeling across photodetector technologies revealed that quantitative CMOS (qCMOS, Hamamatsu Orca-Quest 2) technology offers a slight advantage over conventional EMCCDs and markedly outperforms standard CCDs in this photon-sparse regime. By achieving a sub-electron read noise floor ($\sigma_{\text{read}} \sim$

0.2–0.3 e^- RMS) without active electron multiplication (excess noise factor $F = 1$), qCMOS eliminates the multiplicative noise penalty of EMCCDs ($F \simeq \sqrt{2}$), resulting in a modest boost in absolute SNR. Furthermore, both qCMOS and EMCCD architectures maintain stable performance when optical magnification spreads the PSF across multi-pixel footprints (2×2 and 3×3), with qCMOS *preserving* $> 95\%$ efficiency relative to the Standard Quantum Limit (SQL).

4.2 Perspectives and future directions

Building upon the theoretical modeling and optical system design developed in this work, several key avenues for further research and experimental implementation emerge to advance the performance and scalability of the neutral-atom imaging system.

A critical next step involves conducting a *comprehensive study on field-dependent off-axis aberrations across the complete FoV*. While this work primarily focused on mitigating on-axis spherical aberration, peripheral optical channels in large-scale array geometries inevitably suffer from off-axis astigmatism (Z_5, Z_6), coma (Z_7, Z_8), and angular walk-off. Quantifying how these field-dependent distortions degrade spatial spot symmetry and broaden the effective PSF as a function of pupil filling factor will define the boundaries of the high-fidelity addressable region. Furthermore, *integrating an adaptive filling-factor constraint filter into the automated catalog optimization loop* will allow tailoring the maximum pupil aperture independently for each lens topology, finely balancing geometrical solid-angle collection against higher-order wavefront degradation.

Building on the optimization outcomes, a compelling hybrid optical configuration emerges as the logical culmination of this work. *Combining custom achromatic doublets in the relay stage with a dedicated precision aspheric lens to focus light onto the camera sensor* allows concentrating high optical performance where it is most critical—at the tube lens stage, where marginal rays undergo high refraction angles—while keeping the intermediate relay stage simpler and more cost-effective. Thoroughly modeling this hybrid configuration in Zemax OpticStudio and verifying its performance experimentally will establish an optimized benchmark for high-NA single-atom collection.

To overcome residual static imperfections, *active wavefront correction provides a potential pathway toward diffraction-limited resolution*, though its application to fluorescence requires careful physical consideration. Deploying a Spatial Light Modulator (SLM) or deformable mirror conjugated to the Fourier pupil plane can compensate for global static optical aberrations, such as vacuum window stress or thermal drifts [68, 69]. However, because atomic fluorescence is spatially incoherent across distinct emitter sites, a pupil-plane SLM applies a global phase mask that cannot independently correct position-dependent off-axis aberrations across the array. Moreover, phase modulation relies on mutual spatial coherence across the beam wavefront, requiring further investigation into pupil-plane phase manipulation for incoherent atomic light fields.

On the analytical front, *extending the current SNR framework into a complete statistical readout fidelity model* will refine state discrimination limits. Explicitly modeling the overlap between atom-presence (H_1) and atom-absence (H_0) photon-counting distributions under non-Gaussian noise sources—such as stray light background, sub-Poissonian scatter,

and qCMOS readout noise—will allow deriving optimal thresholding via Neyman-Pearson decision criteria [70]. *Extending this framework to non-destructive, state-selective readout protocols*—where polarization, detuning, and repumping light are tailored to preserve atomic survival across imaging cycles [59]—will enable real-time quantum feedback without relying on state-to-loss detection.

Finally, an immediate experimental benchmark consists of assembling the fluorescence collection setup and testing it against the analytical predictions. *Evaluating the setup using an artificial atom source* (e.g., a single-mode fiber tip or point-like calibration emitter) coupled to the EMCCD and qCMOS camera architectures available in the laboratory will directly validate the collection efficiency, PSF morphology, and sub-electron noise advantages modeled in this work.

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